Uncertainty visualization

CS424: Visualization & Visual Analytics

Fabio Miranda

Slides based on Carolina Veiga's uncertainty class

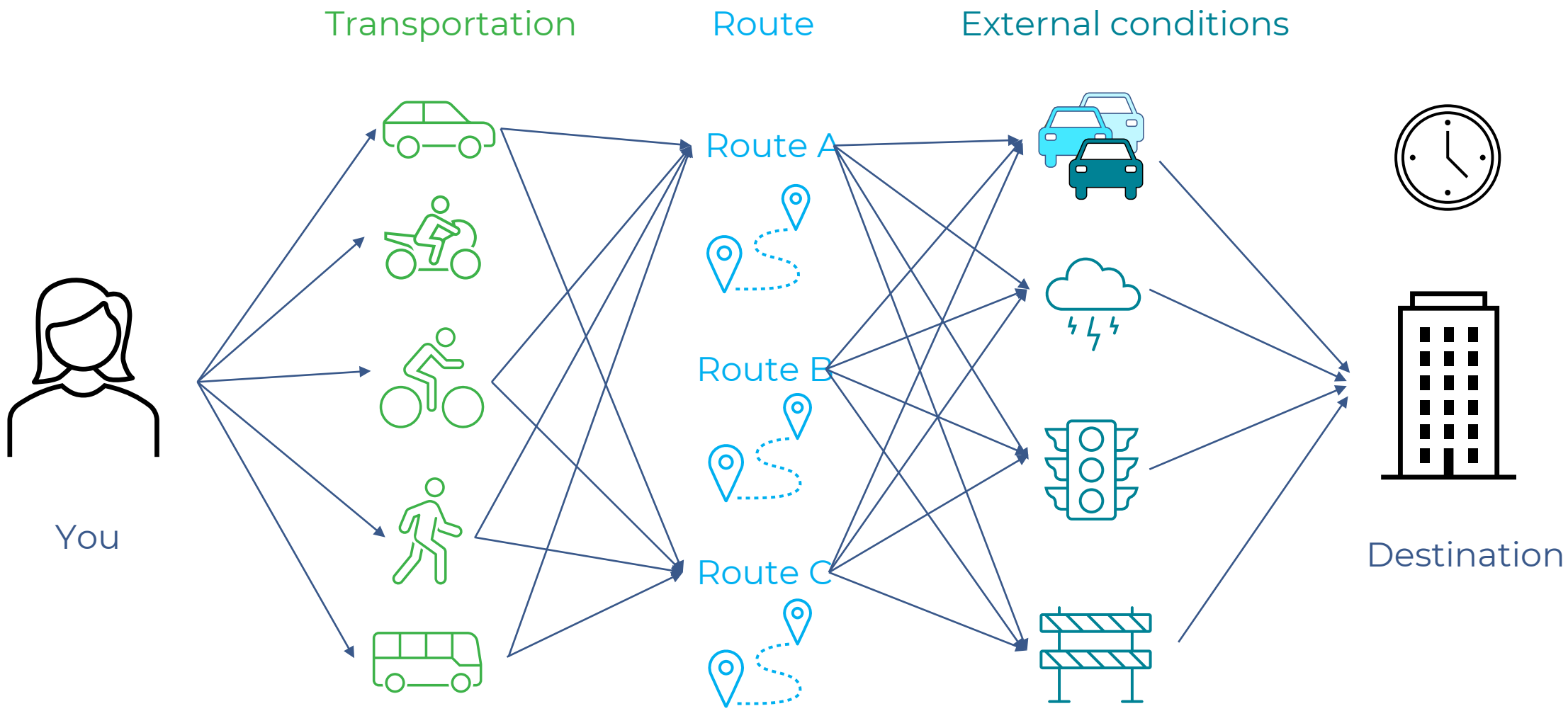




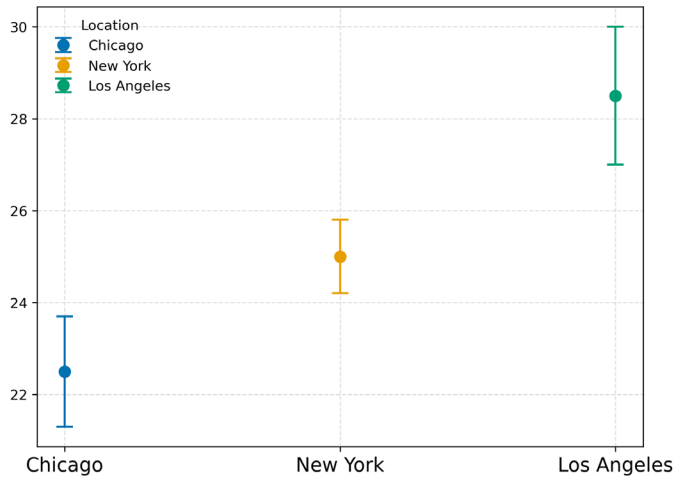


What is uncertainty?

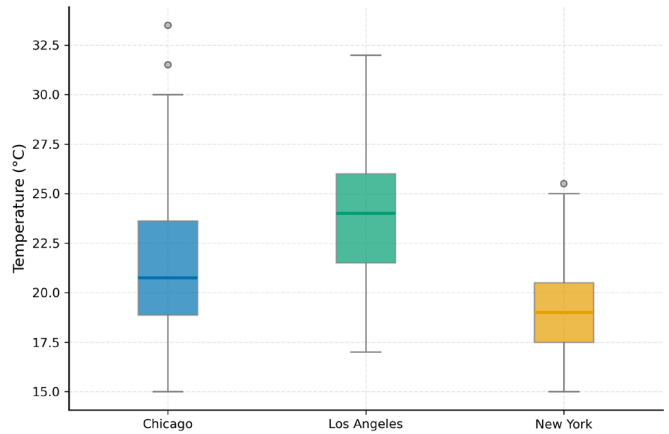
What is uncertainty?



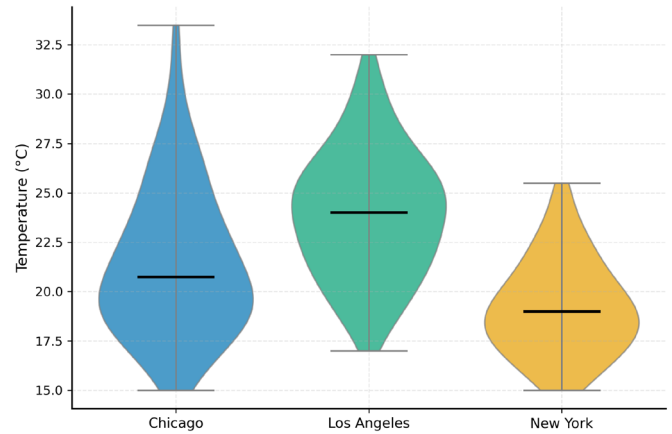
Error Bars



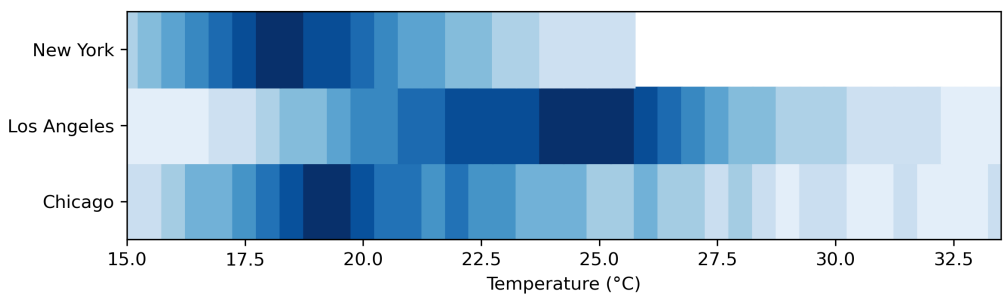
Box Plot



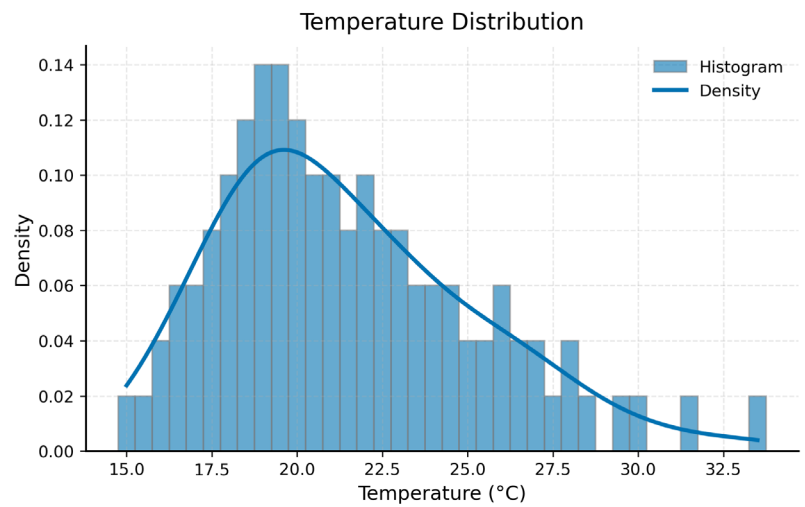
Violin Plot



Gradient Plot



Probability Distribution





Commuting time



Stock market



Criminal investigation



Energy generation



Demand prediction

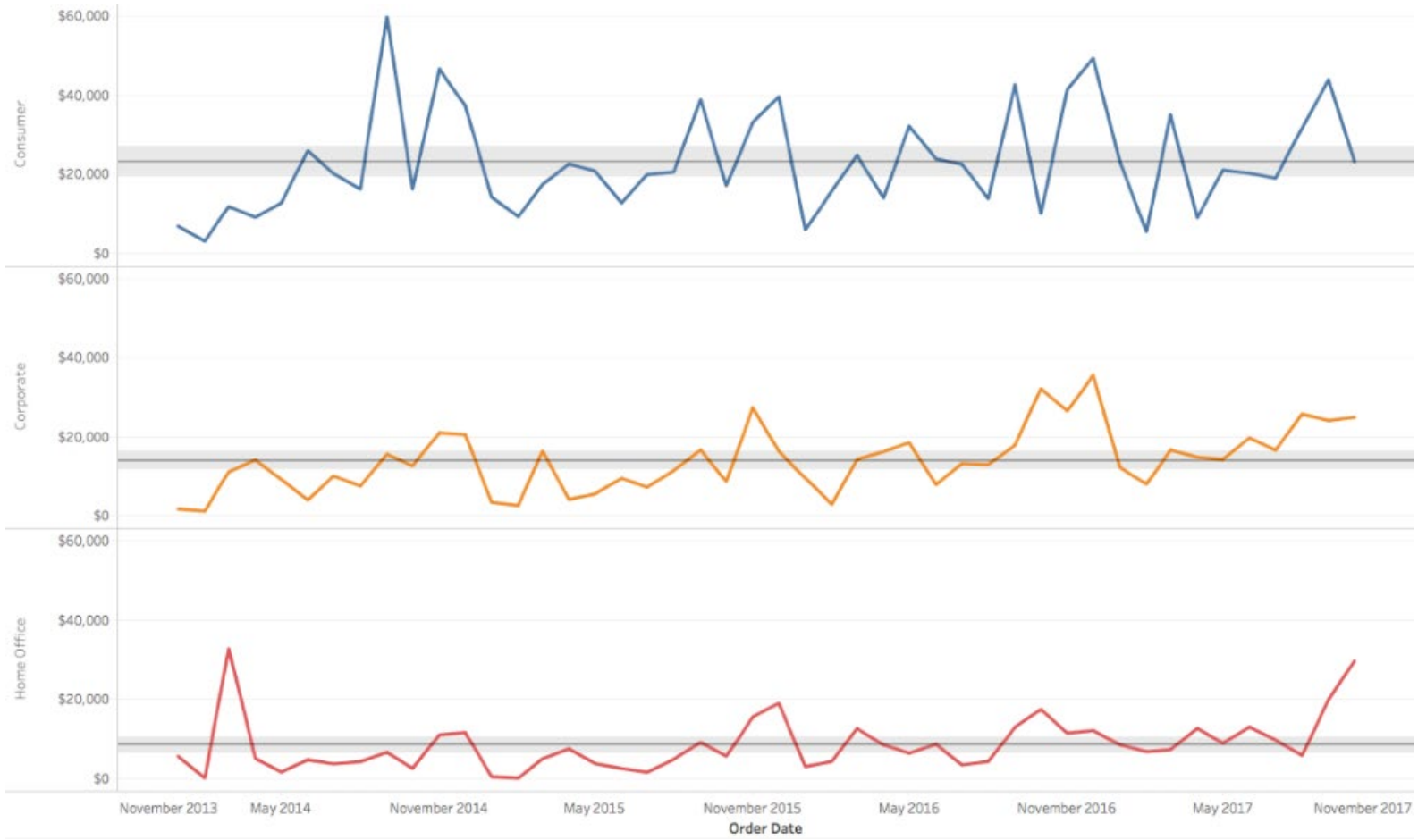


Weather forecasting

Sources of uncertainty in data

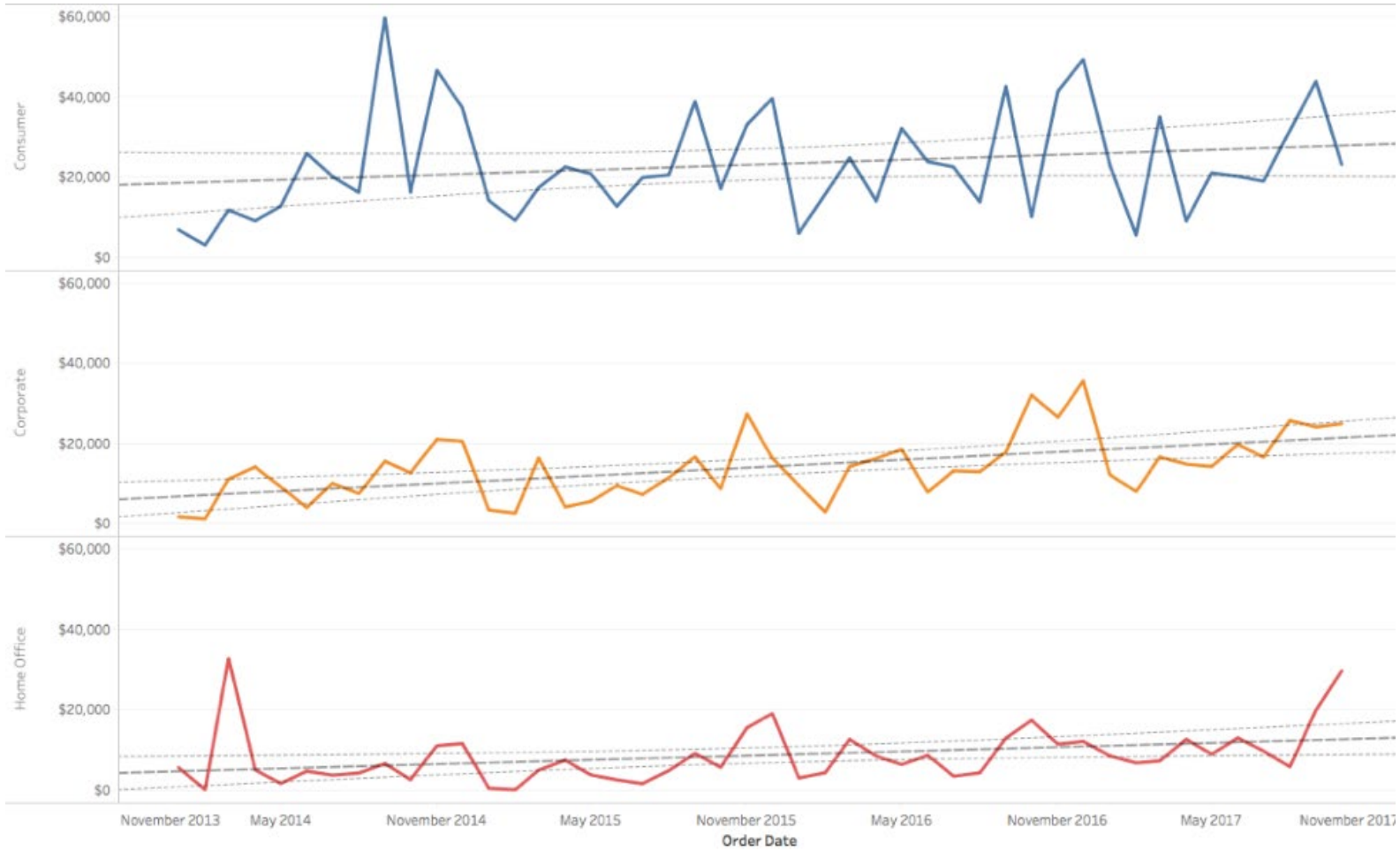
- Measurement: are we sure about the data?
- Forecast uncertainty: what will happen with the data next?
- Model uncertainty: how the data fit together?

Measurement uncertainty



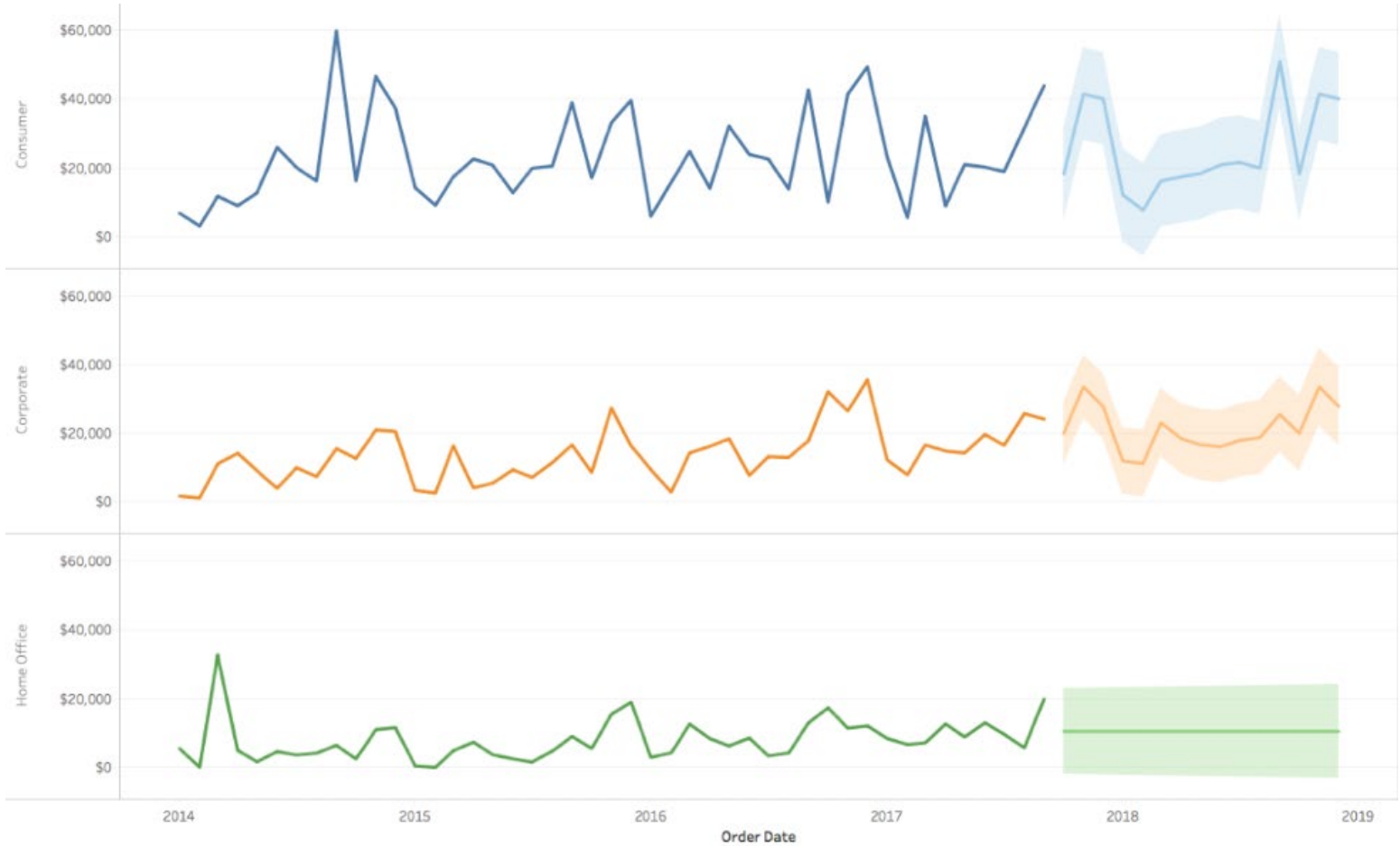
[Correll, Tableau Research]

Model uncertainty



[Correll, Tableau Research]

Forecast uncertainty



[Correll, Tableau Research]

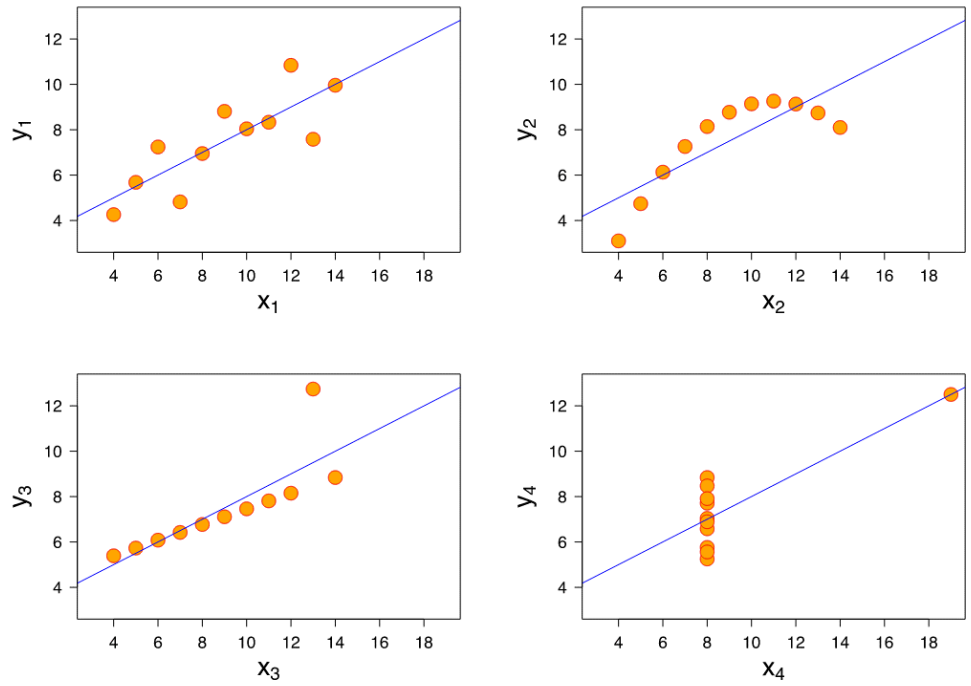
What does uncertainty mean?

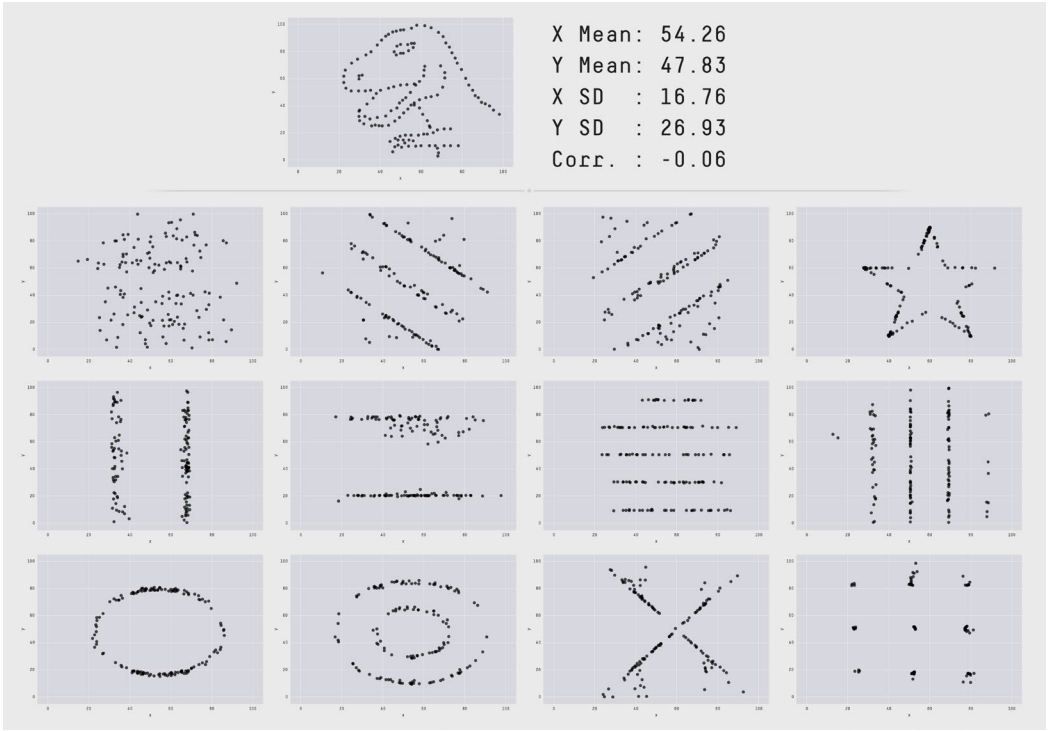
"Any one of a number of potentially interconnected quantitative, qualitative, or factors that affect the quality, reliability, or utility of your data or data-driven decisions. Anything that can cause you to be unsure about your data or how to use it."

Michael Correll
Tableau Research

Data aggregation generates uncertainty

- Same stats, different graphs [Matejka and Fitzmaurice, 2017]





Summary statistics

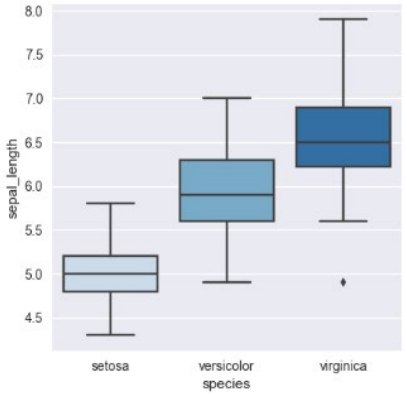
- Summarizes statistical information
- Measures of locality:
  - Mean
  - Median
  - Interquartile mean
- Measure of spread:
  - Standard deviation
  - Range
  - Variance
  - Absolute deviation
  - Interquartile range

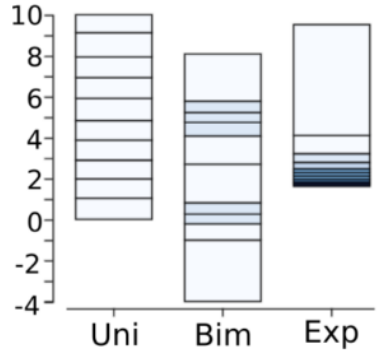
Uncertainty visualization

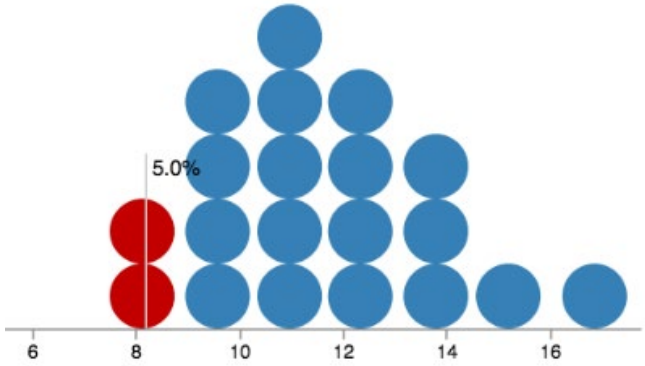
- When you have a whole range or distribution of numbers, avoid visualizing single numbers.
- Graphical annotations of distributional properties
  - Histograms and density plots
  - Intervals and ratios
  - Distributions
- Visual encodings of uncertainty
- Hybrid approaches

Possible solution: quantiles

- Robust statistics with clear interpretation







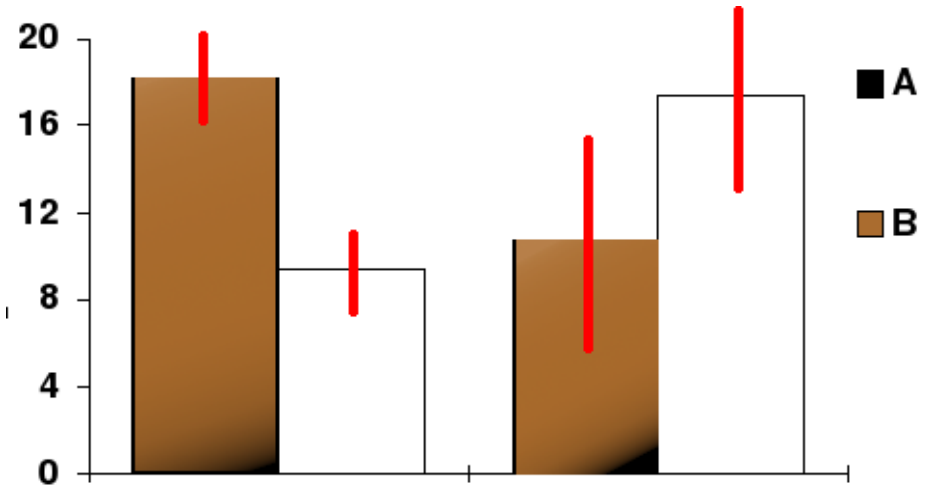
- Important: quantiles cannot be merged

$$Mean(A \cup B) = \frac{(\sum A + \sum B)}{(|A| + |B|)}$$

$Median(A \cup B)$ is not a function of $Median(A)$ and $Median(B)$

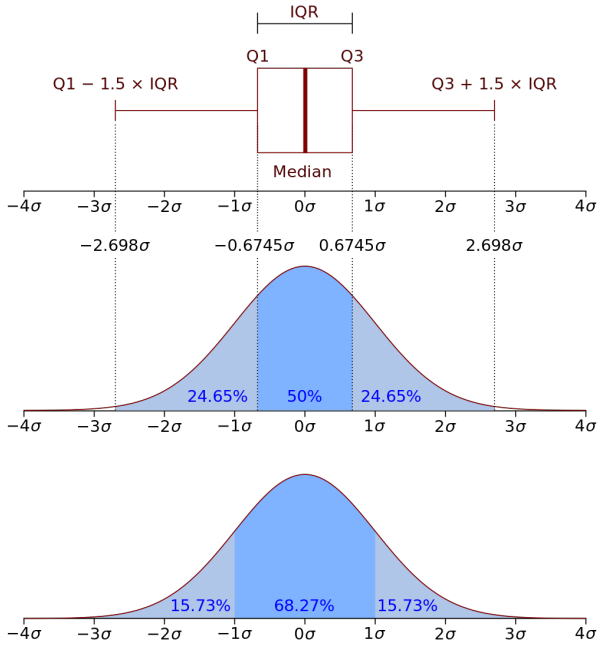
Intervals and ratios

- Error bars: show variability of data, indicating error or uncertainty.



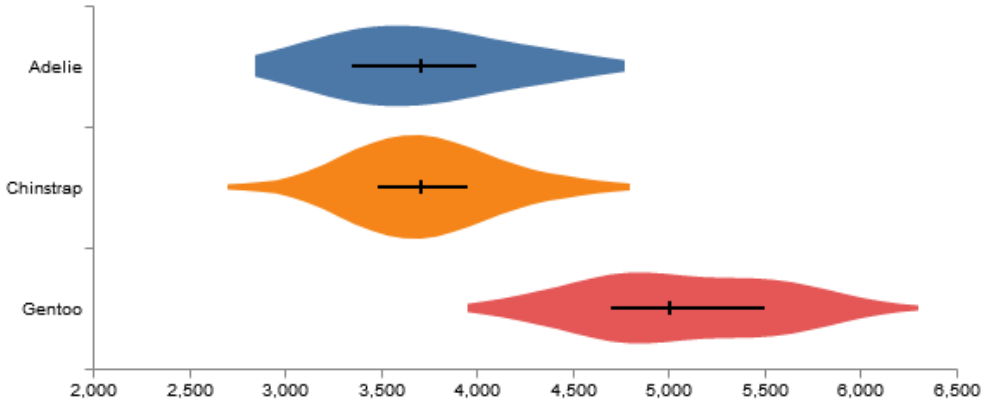
$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i + \mu)^2}$$

Error bars: show variability of data, indicating error or uncertainty.

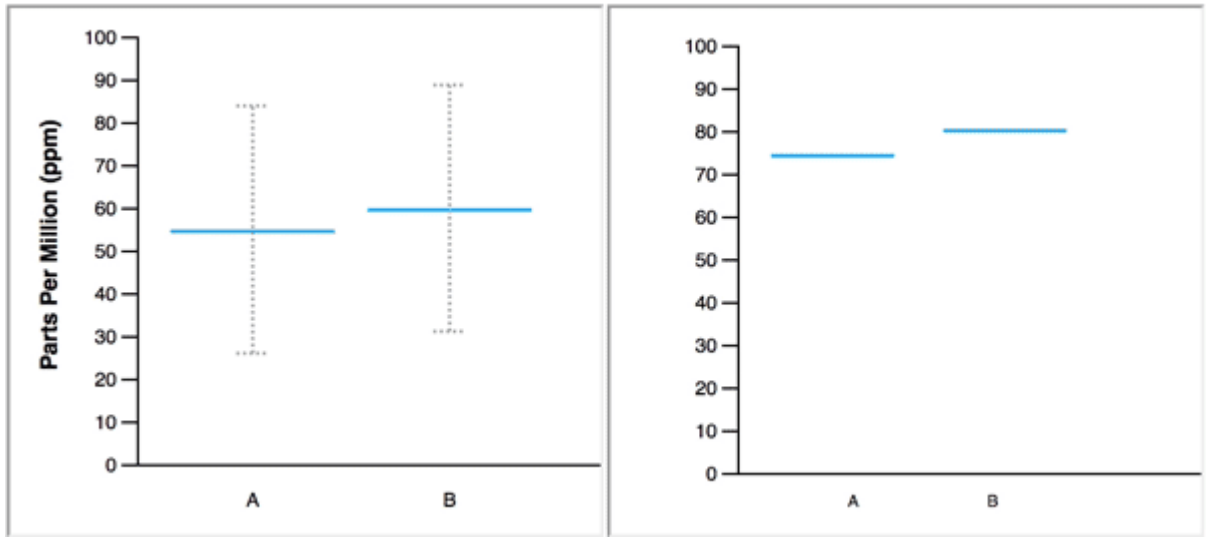


Distributions

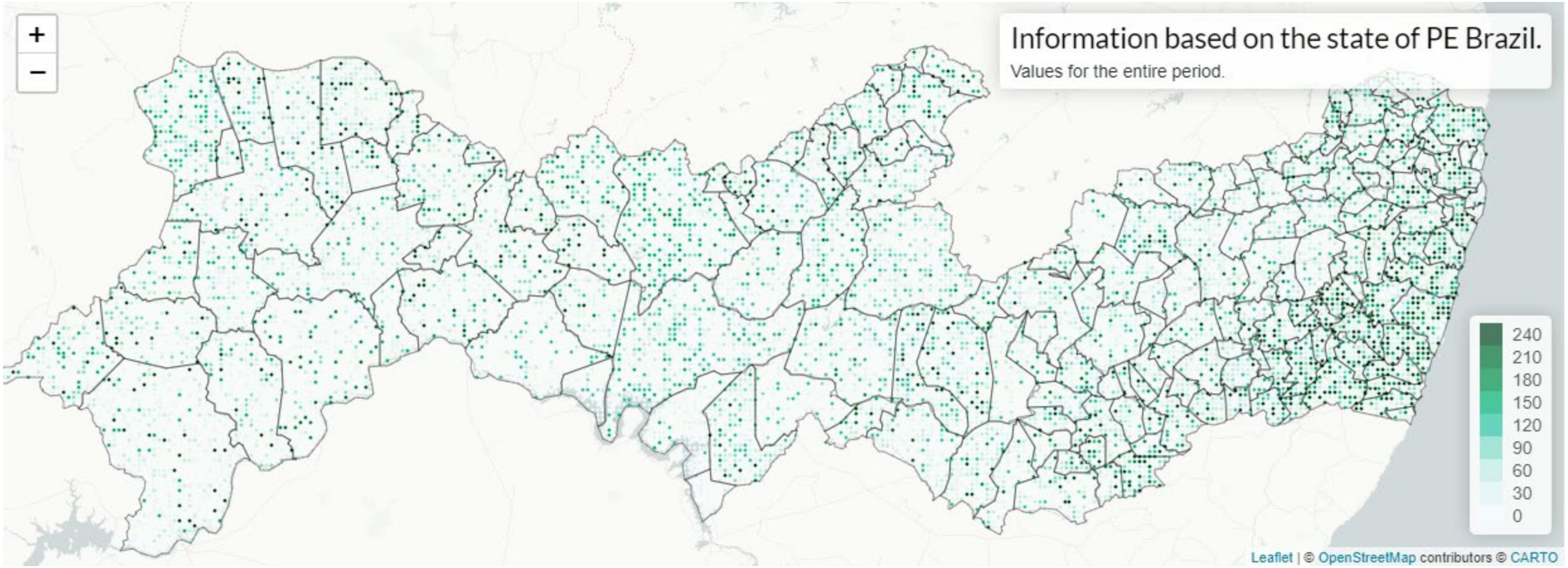
- Violin plot: similar to box plots, but show probability density at different values.



Hypothetical outcome plots: visualize a set of draws from a distribution.



Quantile dot plots

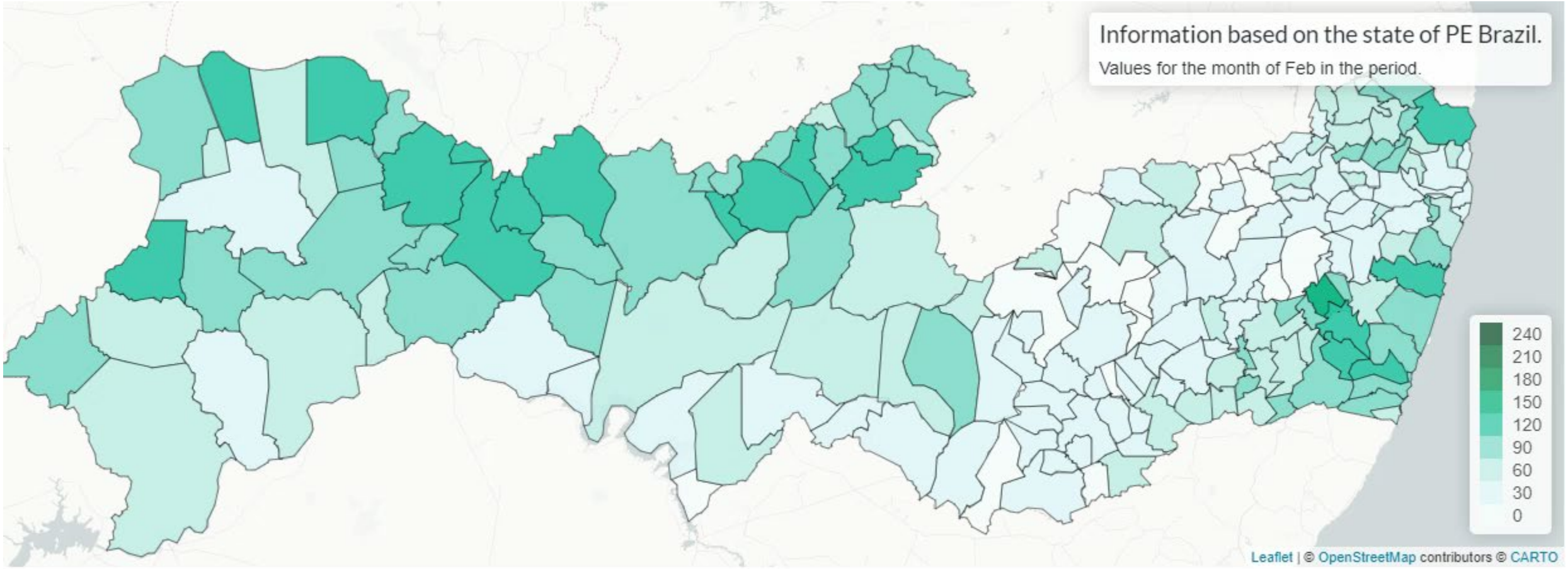


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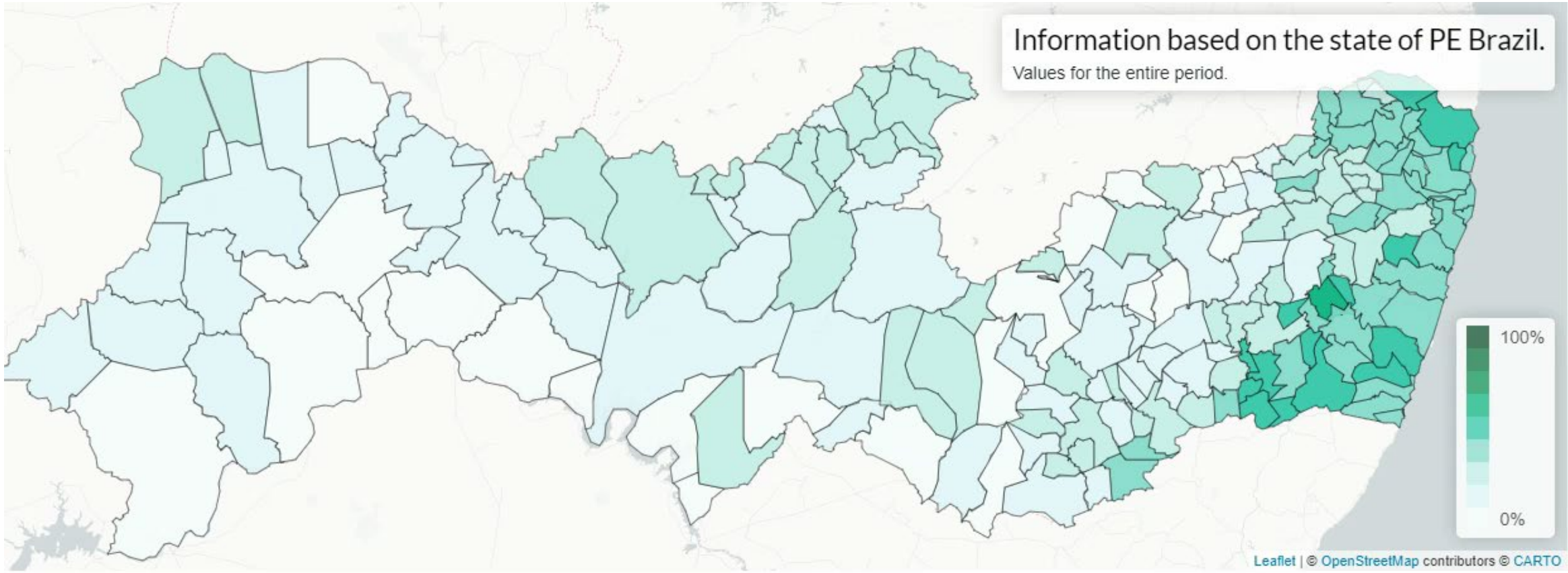
Hypothetical outcome plots

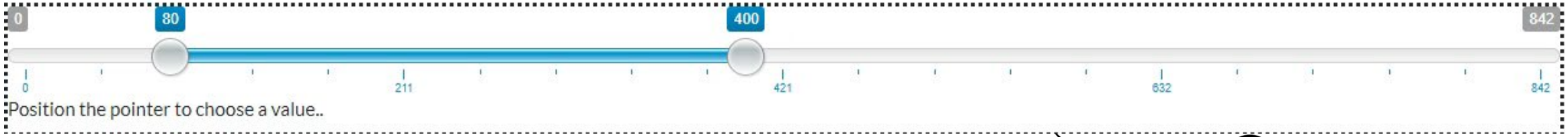


Animate Stop

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Interaction maps



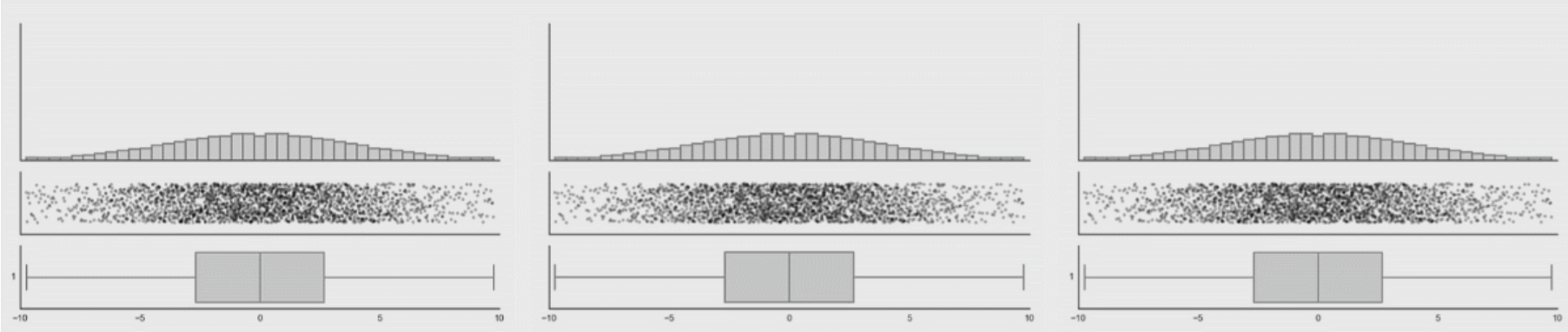


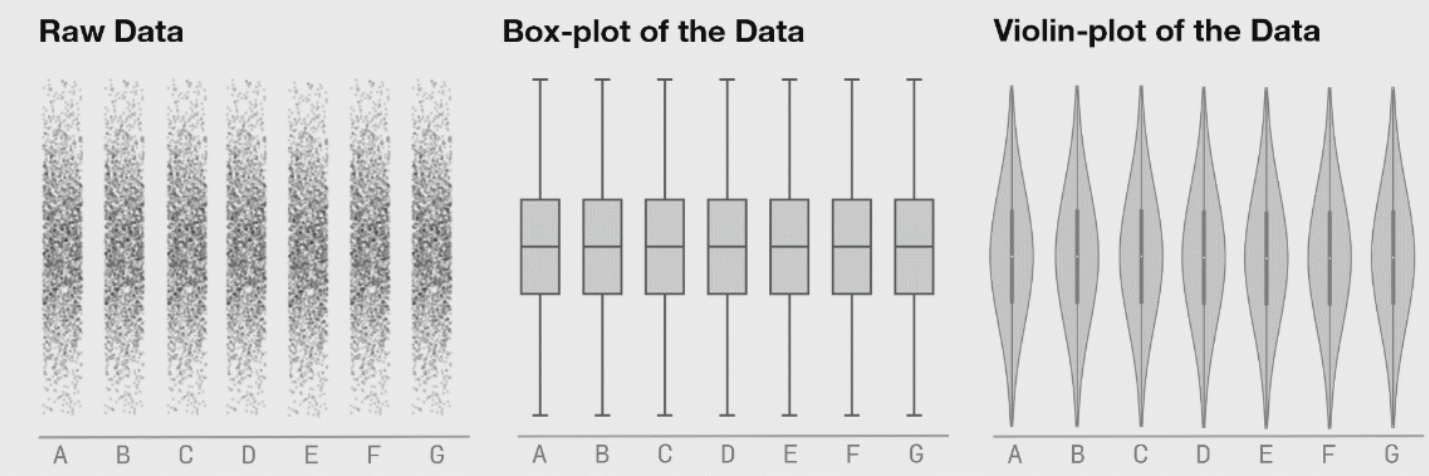
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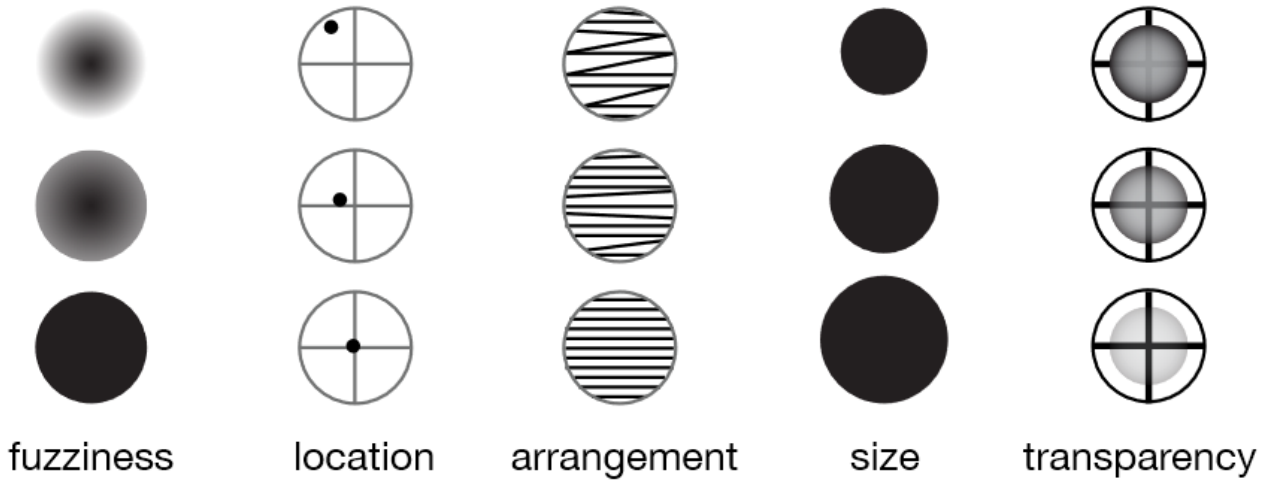
Data aggregation generates uncertainty





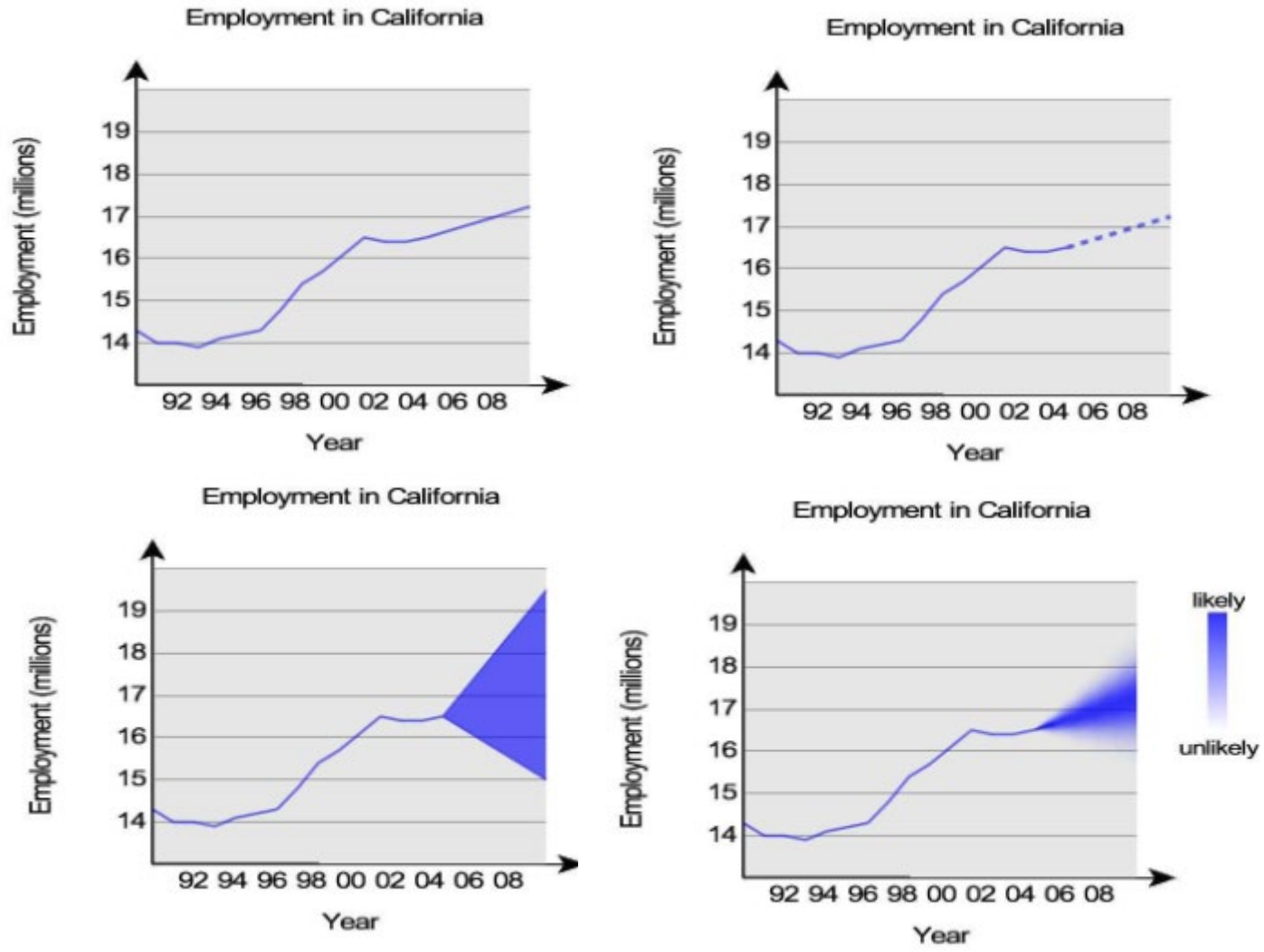
Visual encodings of uncertainty

- Additional visual channels can be used to characterize the uncertainty: hue, texture, opacity
  - Fuzziness
  - Location
  - Arrangement
  - Size transparency



[Padilla et al., 2020]

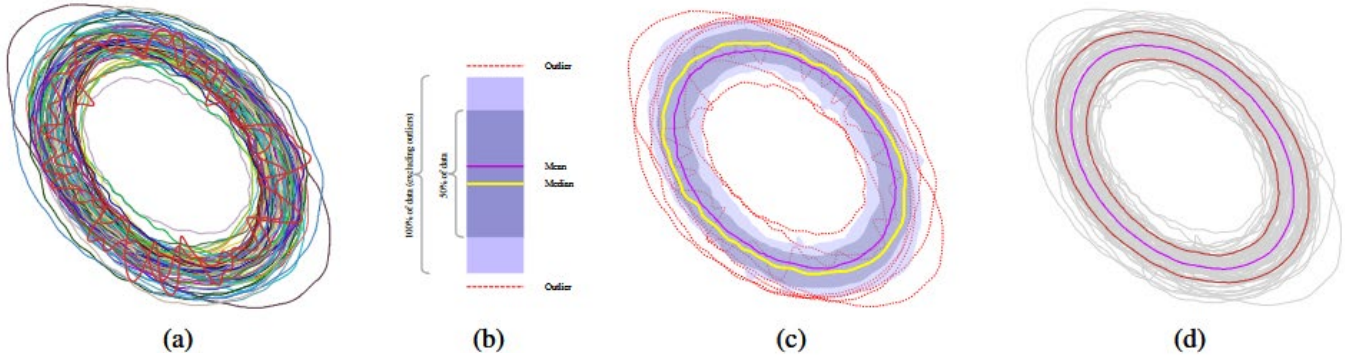
Visual encodings of uncertainty

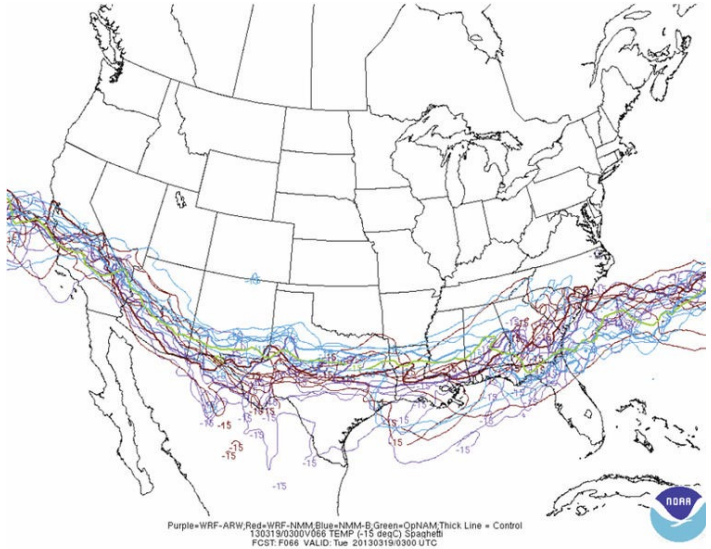


[Streit et al., 2007]

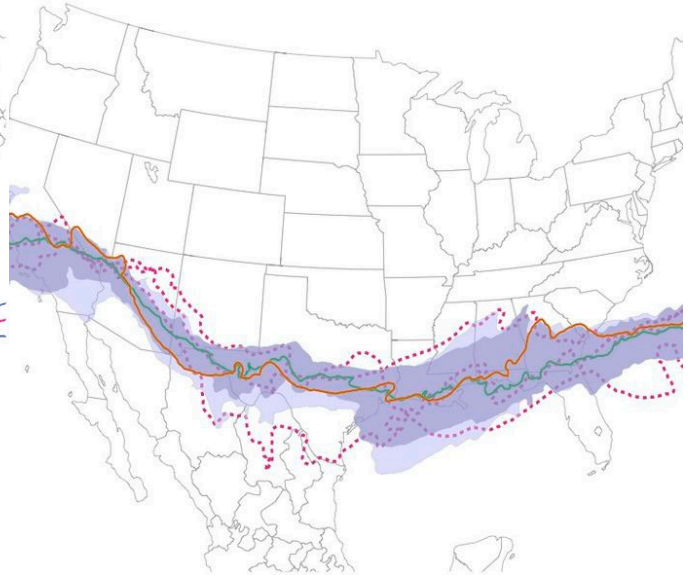
Hybrid approaches

- Contour boxplot:









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ggdist

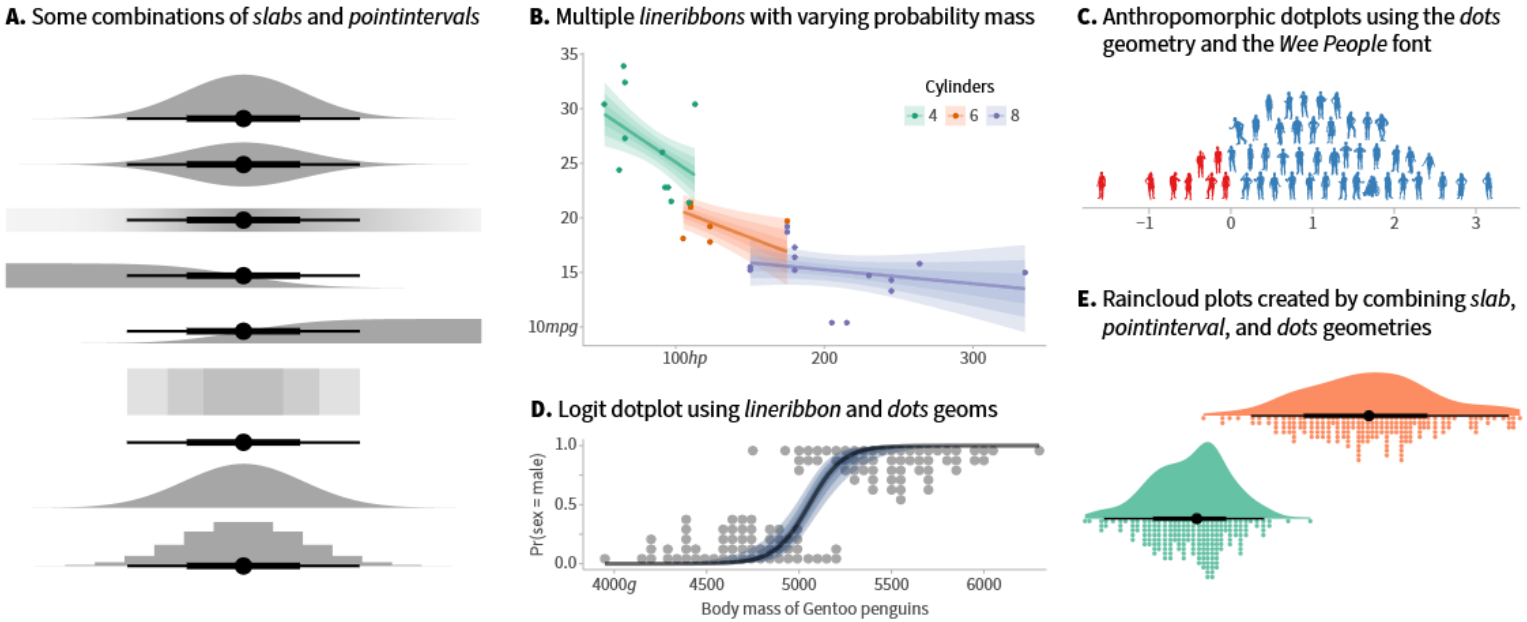


Fig. 1: Examples from the three major classes of geometries in *ggdist*: **(A)** slabintervals, such as density plots, CDFs, intervals, and gradient plots; **(B)** lineribbons, such as uncertainty bands and fan charts; and **(C)** dots, such as dotplots and beeswarm charts. Myriad combinations of these are possible, leading to charts like **(D)** *logit dotplots* and **(E)** *raincloud plots* [1].

ggdist

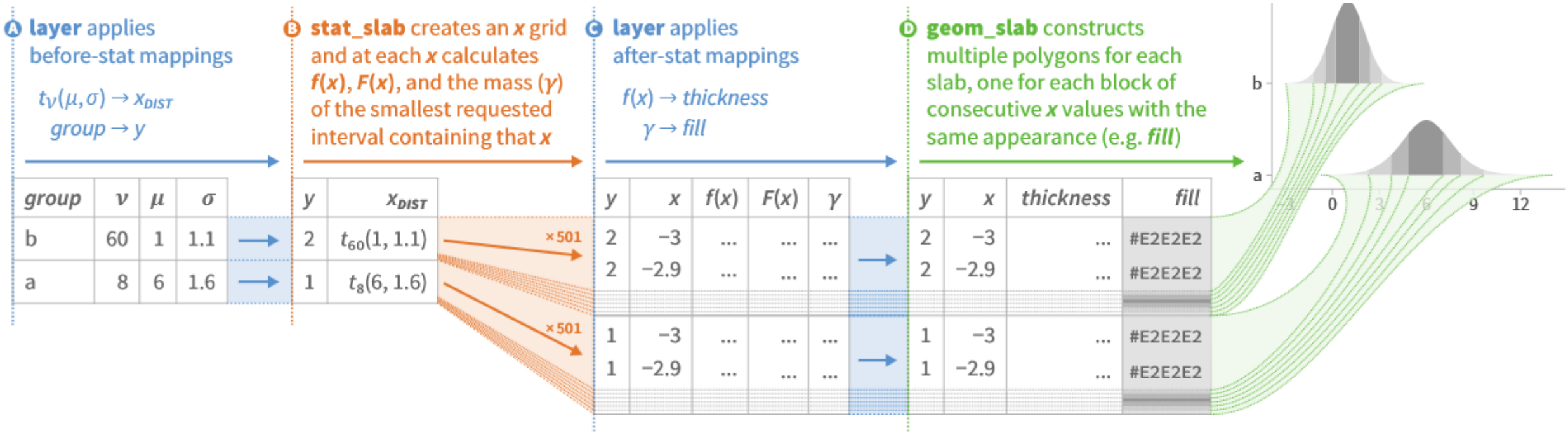
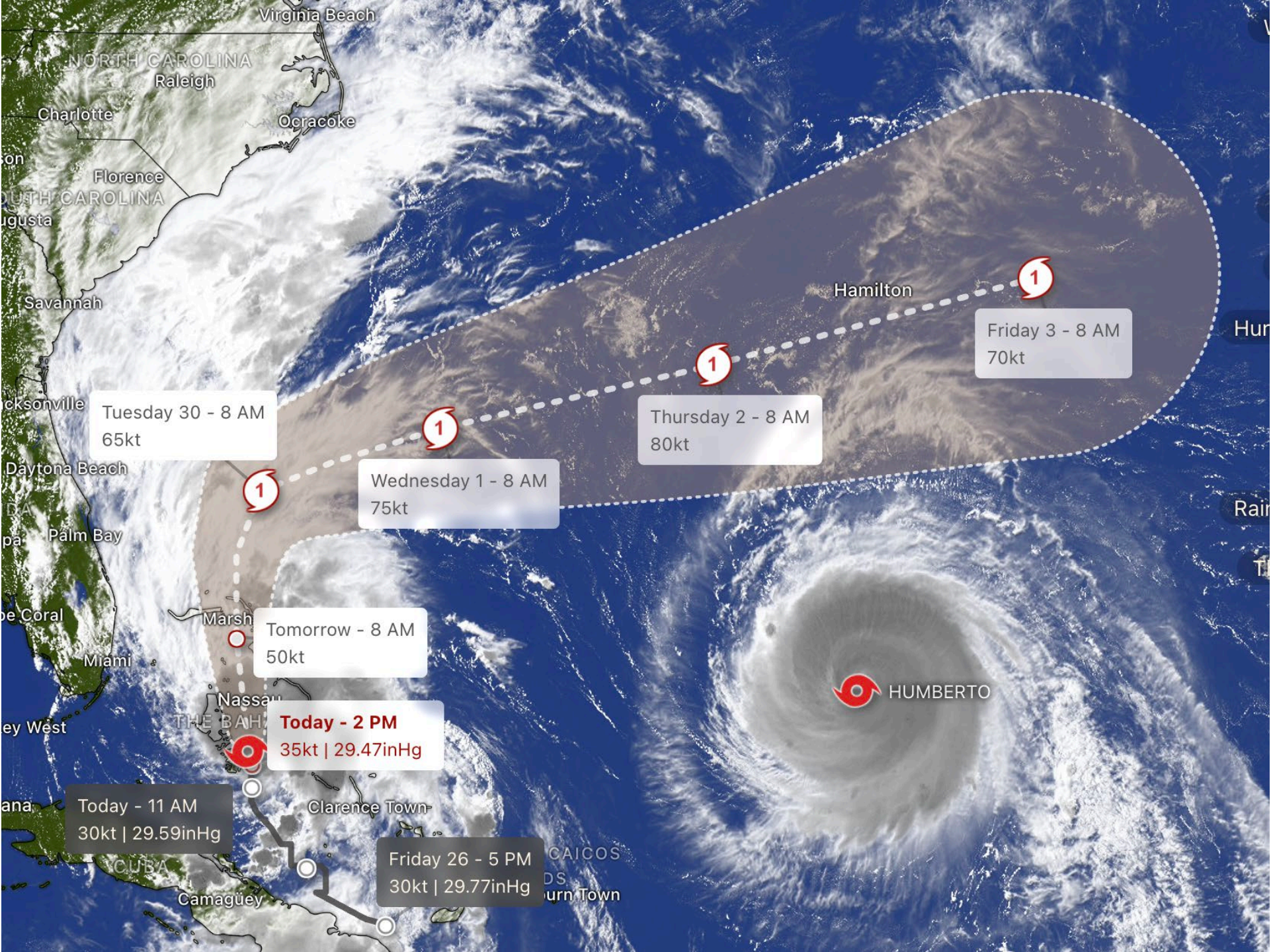
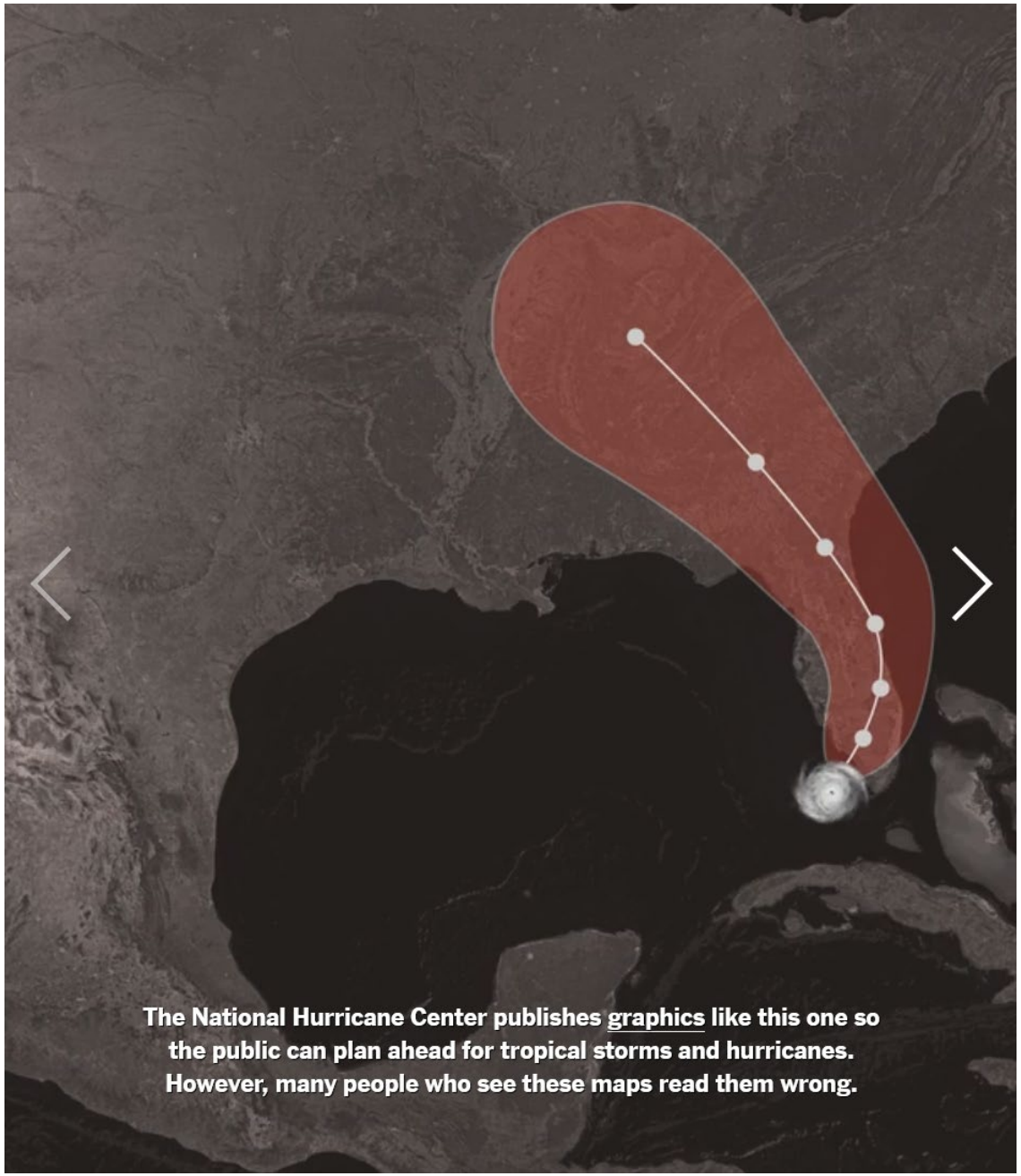


Fig. 2: Flow of data through the visualization construction pipeline for a Helske [18]-style interval+density plot using `ggdist::stat_slab`. The `layer`, part of *ggplot2*, applies initial aesthetic mappings, translating input data into columns understood by `stat_slab`, which then constructs a larger data frame containing evaluations of distribution functions over an x grid for each distribution (the user can specify the grid size if needed; the default size is 501). Then, `layer` applies a second set of aesthetic mappings, and `geom_slab` uses the final data frame to construct the plot.

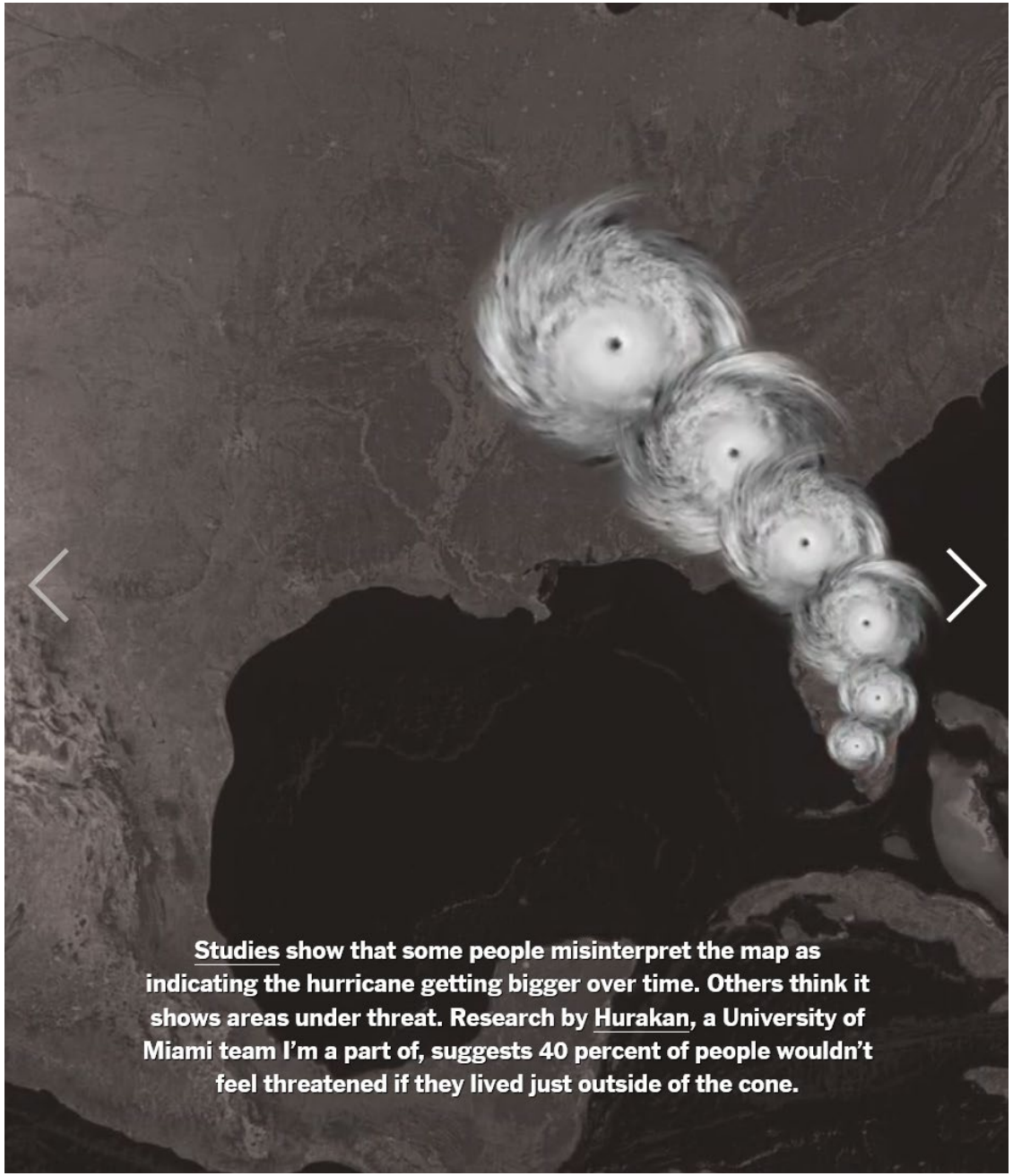


[Windy: Wind map & Weather forecast](https://www.windy.com/)

https://www.windy.com/

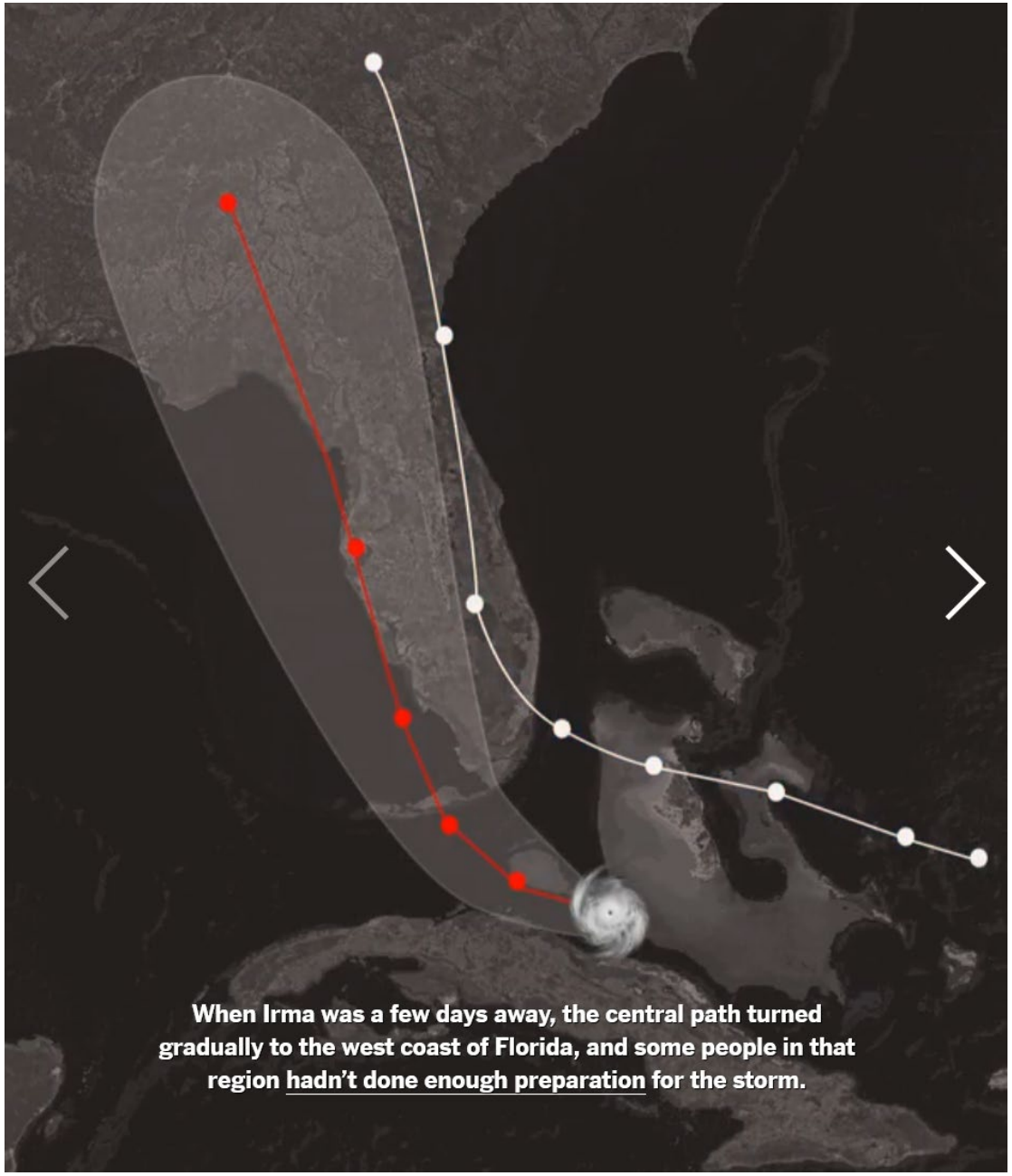


The National Hurricane Center publishes graphics like this one so the public can plan ahead for tropical storms and hurricanes. However, many people who see these maps read them wrong.



Studies show that some people misinterpret the map as indicating the hurricane getting bigger over time. Others think it shows areas under threat. Research by Hurakan, a University of Miami team I'm a part of, suggests 40 percent of people wouldn't feel threatened if they lived just outside of the cone.

Source: [NY Times]





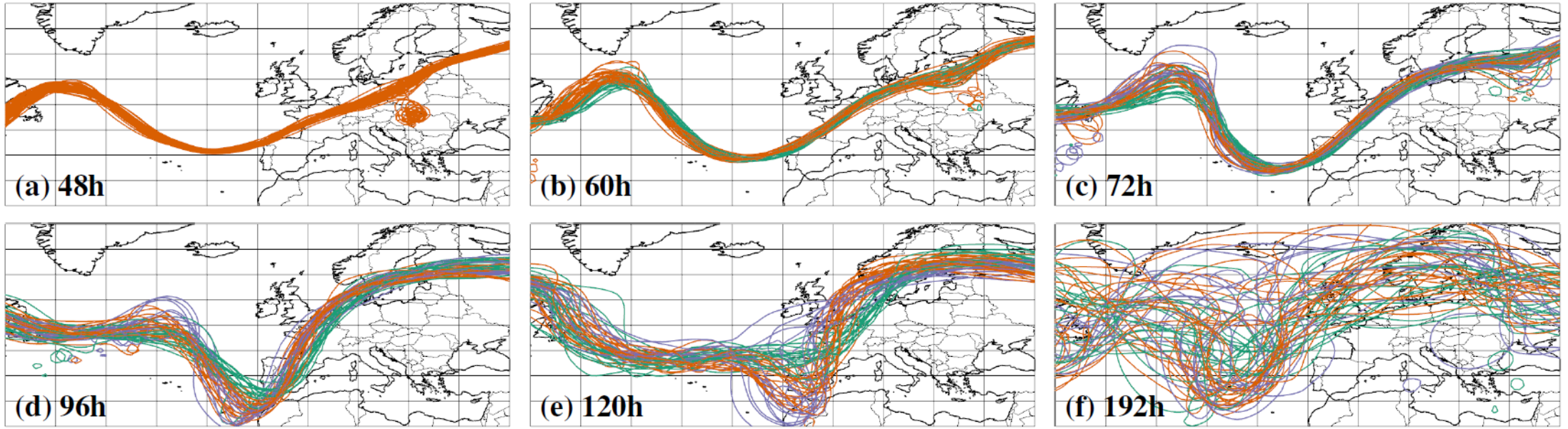
Source: [NY Times]



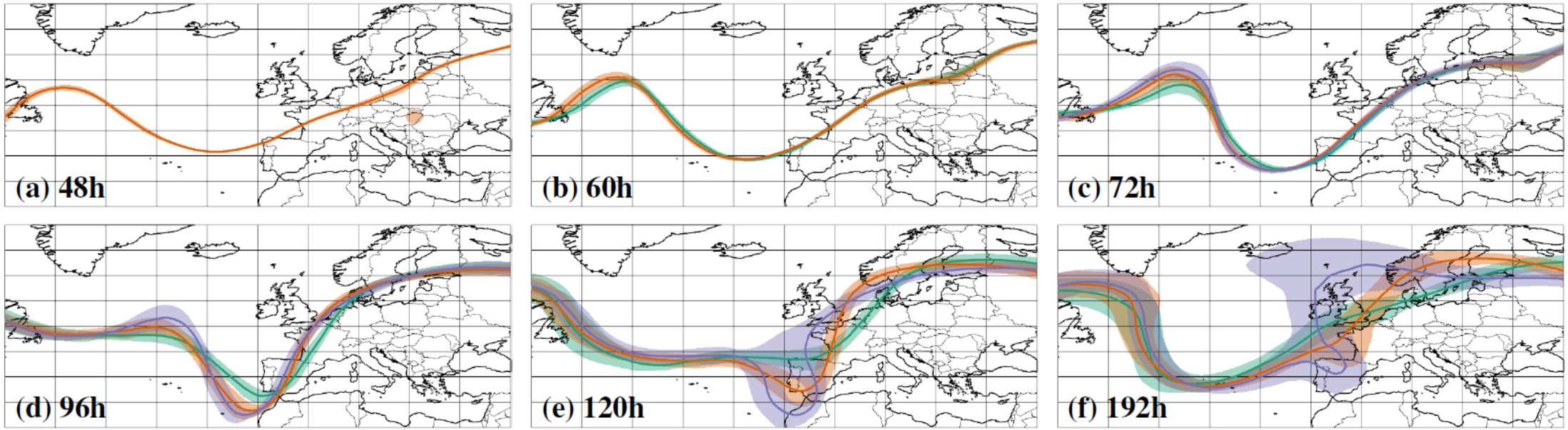


Source: [NY Times]

Spaghetti

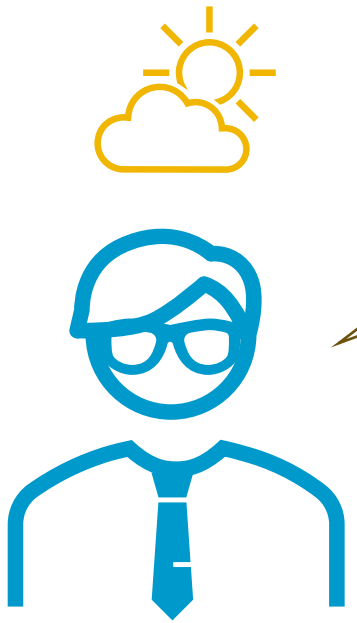


Contour boxplots



Ferst et al. 2017

We have an **ensemble**, and need to answer:
How probable is a storm in this area?
Which setup gives the best results?
How do different setups affect the results?



Climate Expert

How can I help?



Data Visualization Expert

I am used to static images

Creating some interactive maps and charts.

Creating a dashboard.

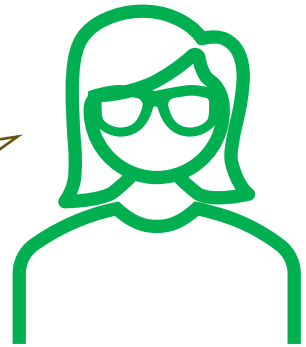
Plotting variables in a 3D map?

A common dashboard may not be enough.



Climate Expert

Show me your data.
Tell me about your pipeline.
How do you usually work?



Data Visualization Expert

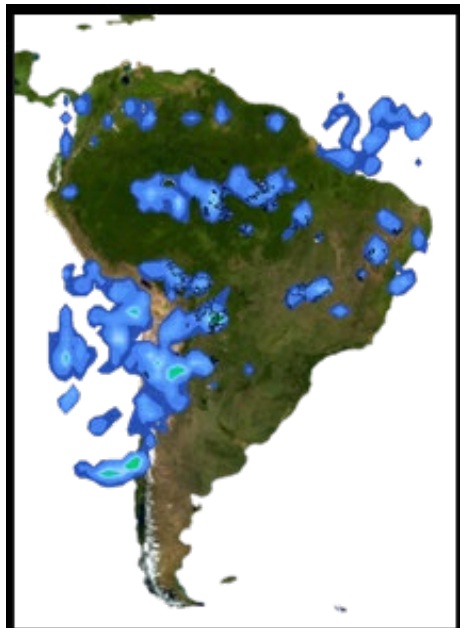
How to model the atmosphere?

Start

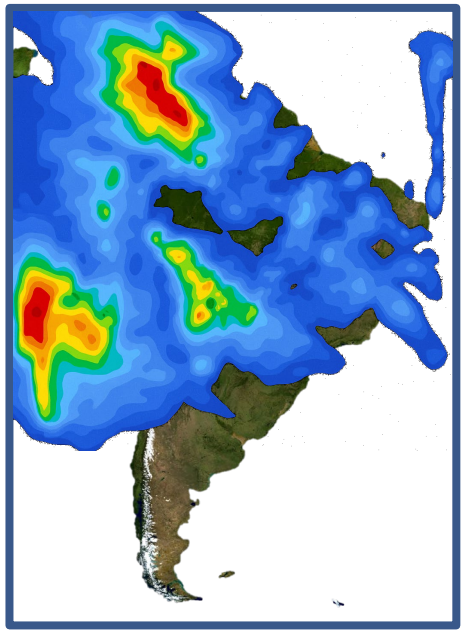
End



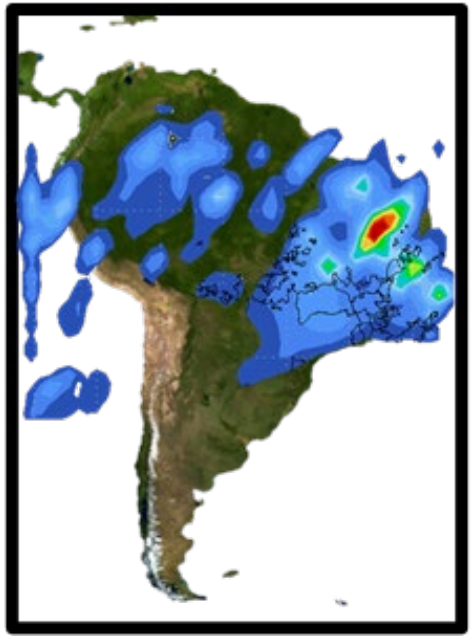
Time step 1



Time step 2



Time step 3

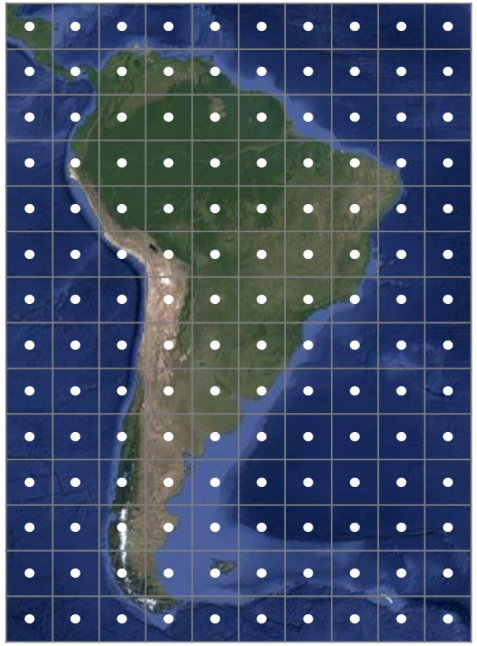


Time step 4

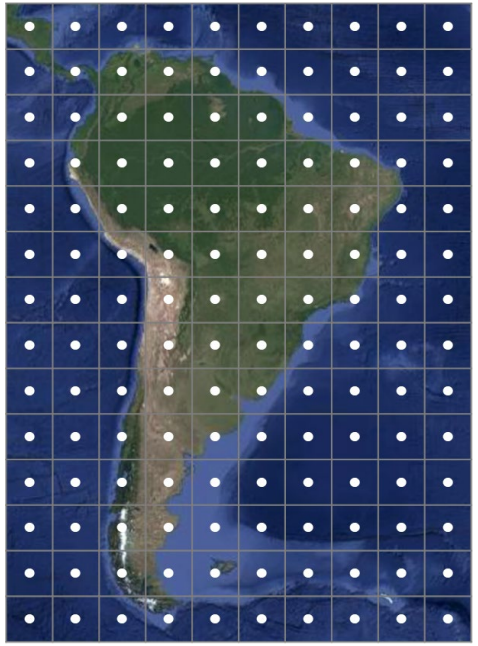
How to model the atmosphere?

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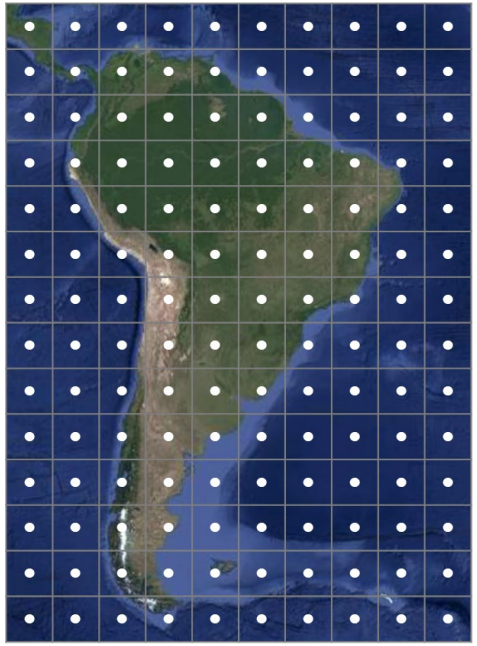
End



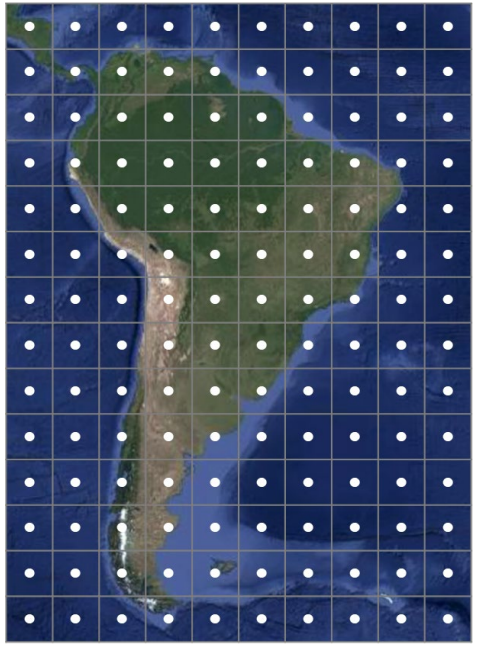
Time step 1



Time step 2



Time step 3



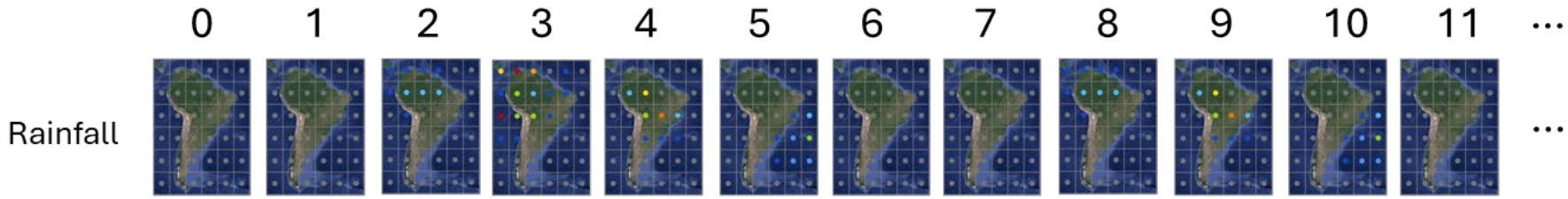
Time step 4

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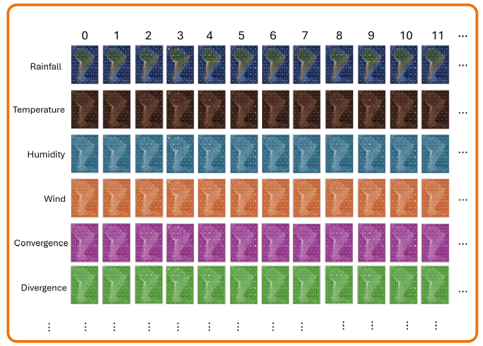
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VARIABLES

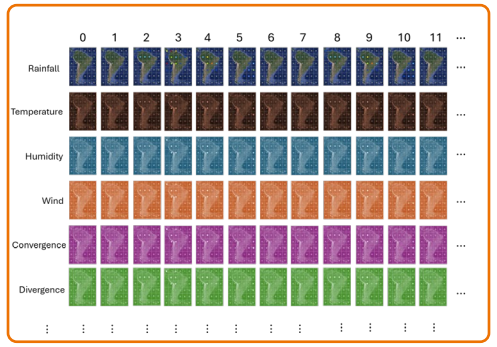


RUNS

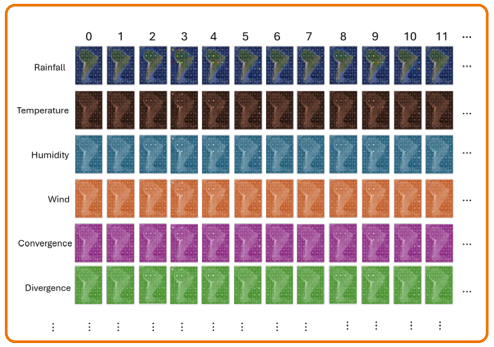
Run 1



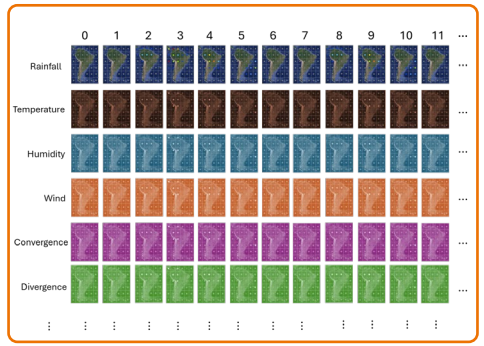
Run 2



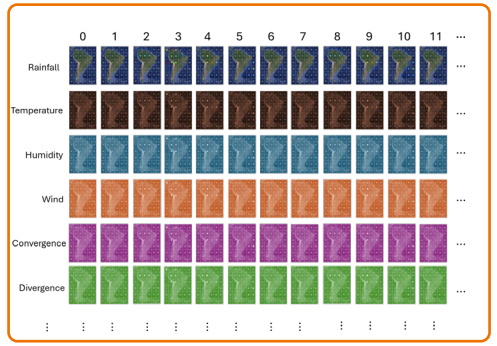
Run 3



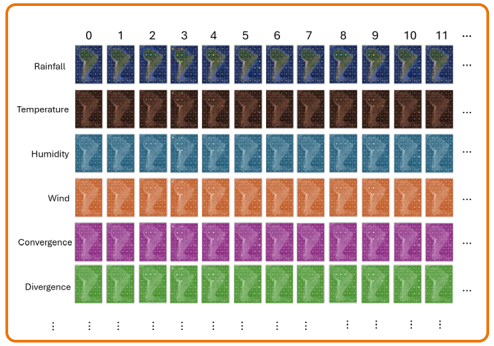
Run 4



Run 5



Run 6



...

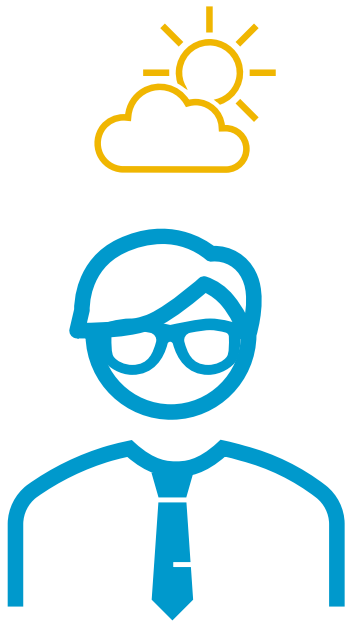
Ensemble

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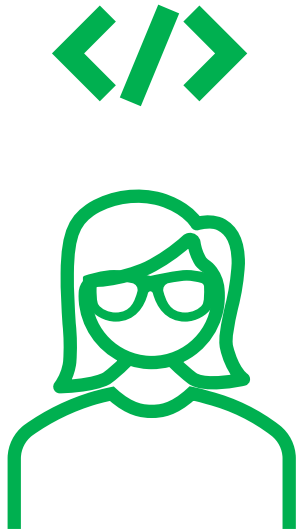
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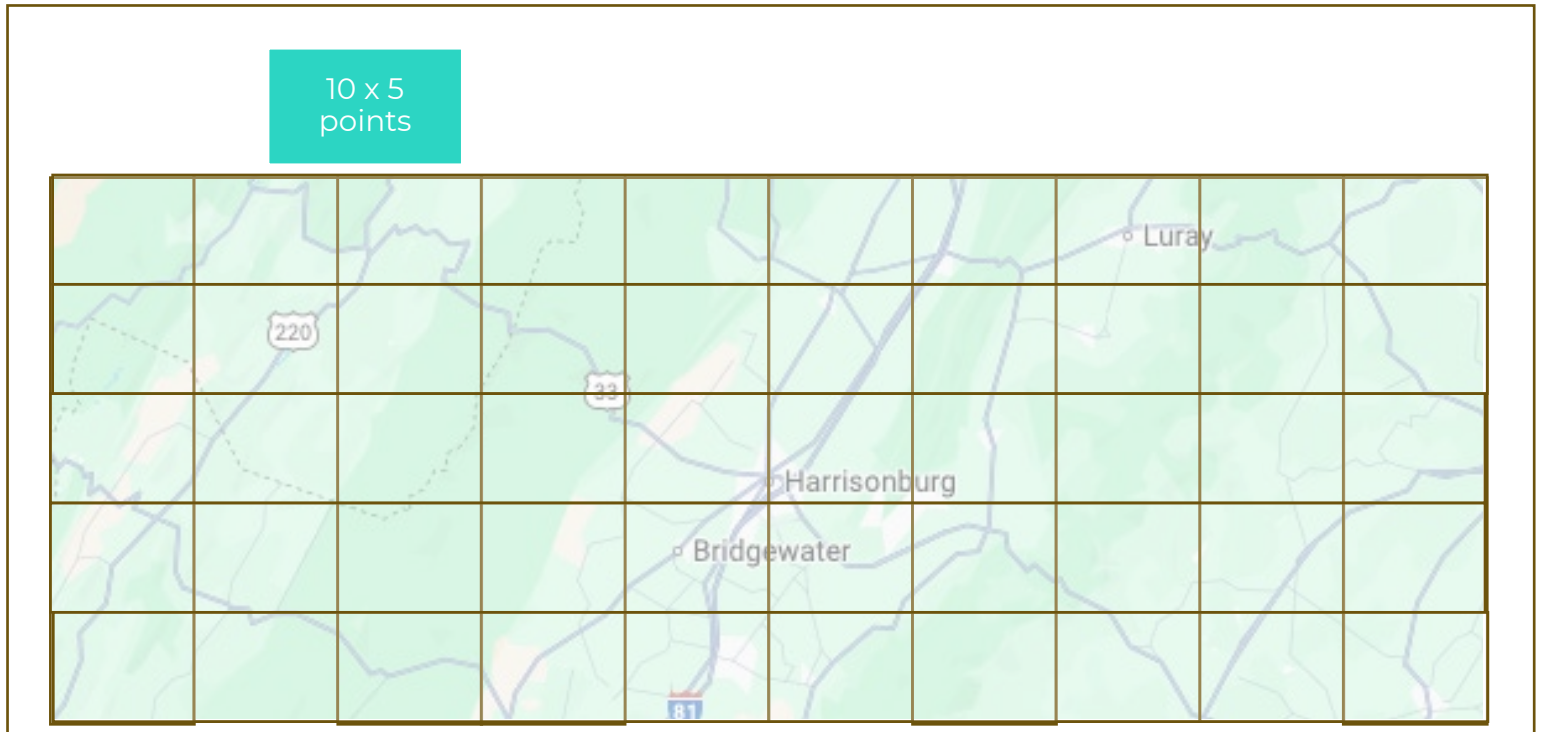
Tell me more.

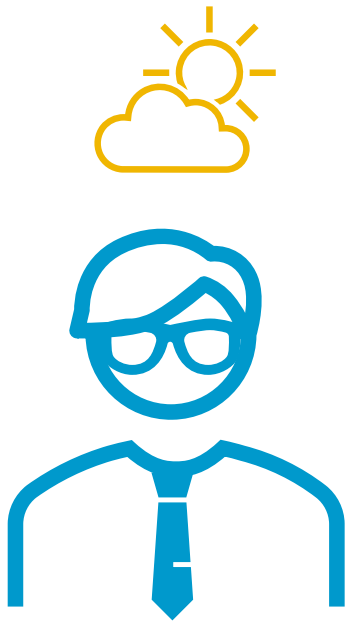


Climate Expert



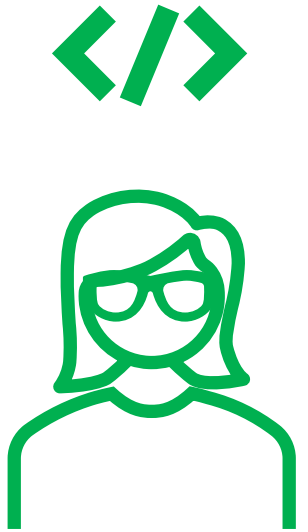
Data Visualization Expert





Climate Expert



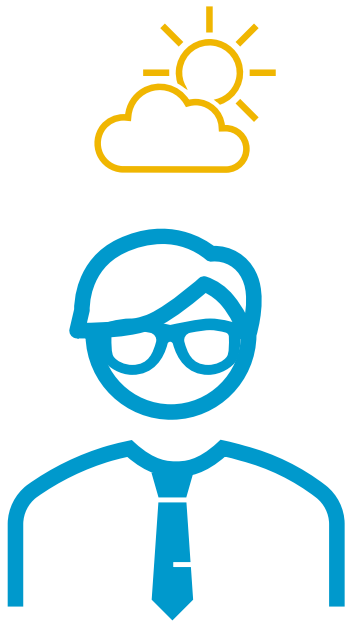


Data Visualization Expert

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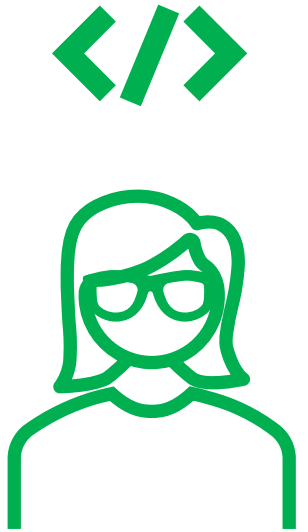


Climate Expert

10 x 5 points | 12 hours of forecasting | Rain | 10 runs

	t1	t2	t12
Run 1			...
Run 2			...
⋮	⋮	⋮	⋮
Run 10			...

10 x 5 = 50
x 12 = 600



Data Visualization Expert

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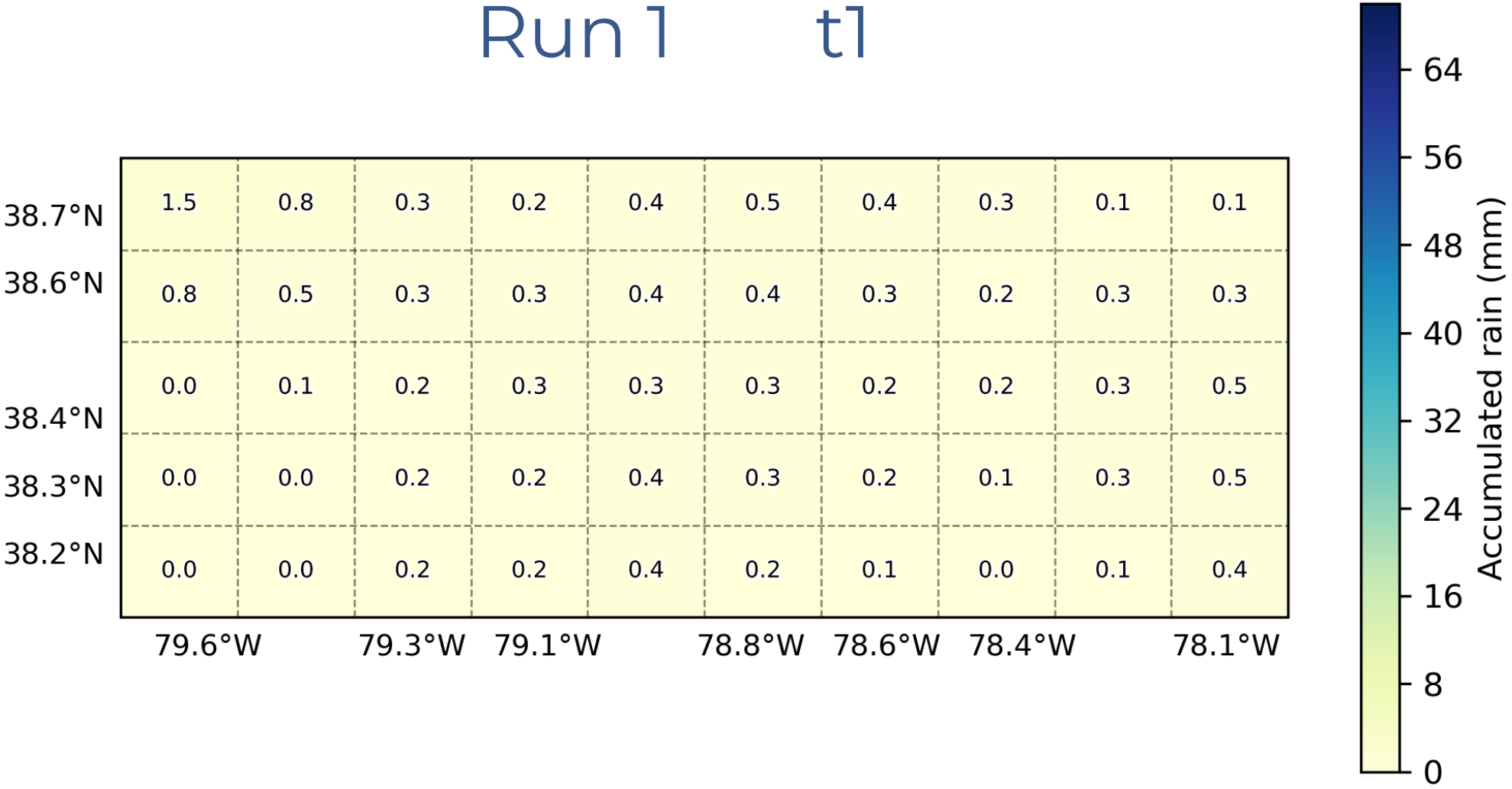
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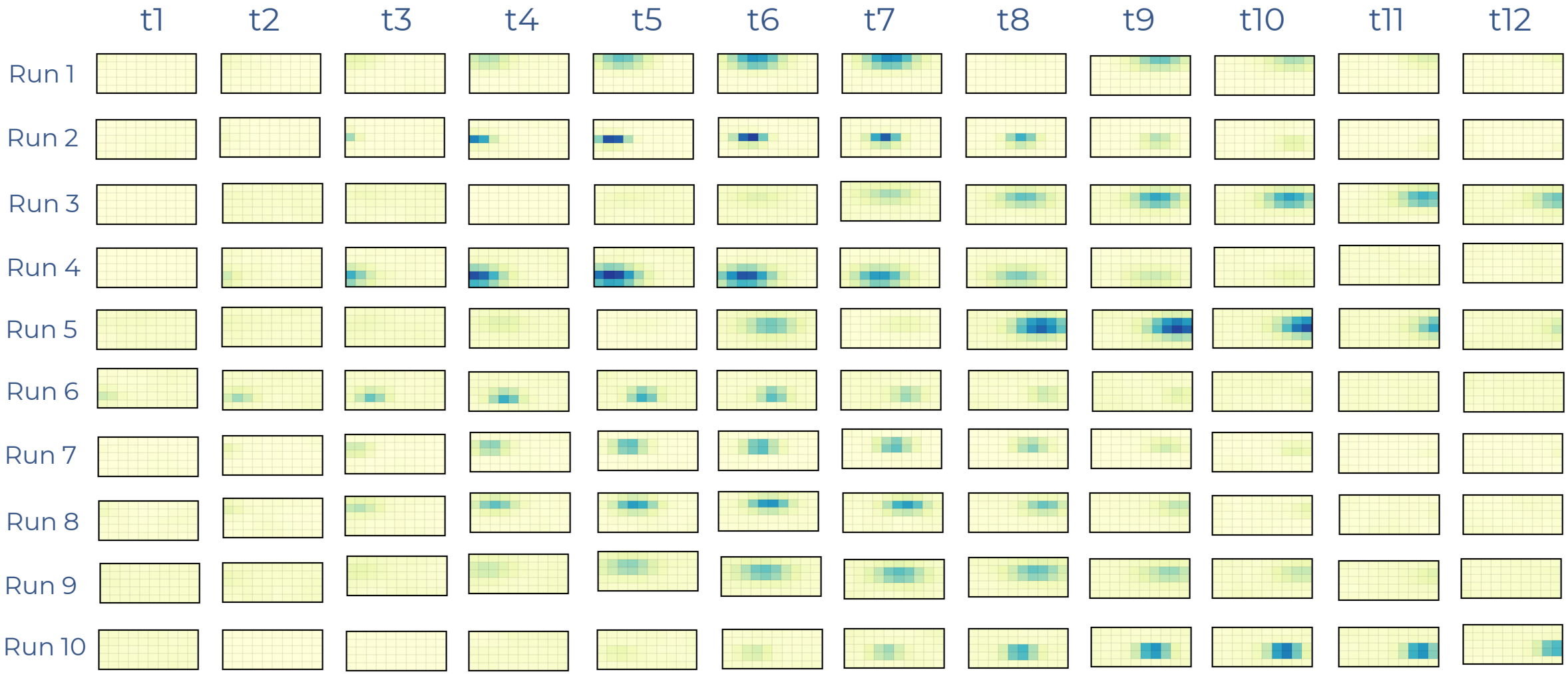
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What’s the probability of rain?

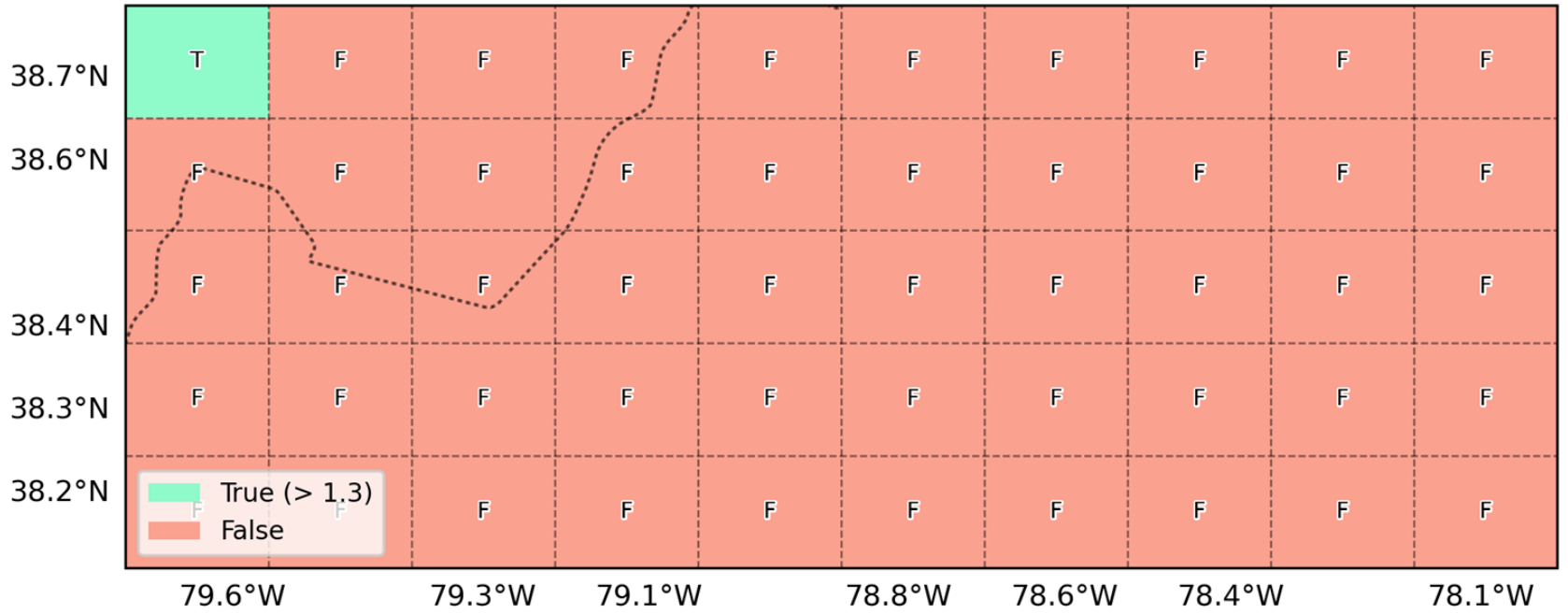
How much? Where?

Run 1 t1


38.7°N
38.6°N
38.4°N
38.3°N
38.2°N
1.5 0.8 0.3 0.2 0.4 0.5 0.4 0.3 0.1 0.1
0.8 0.5 0.3 0.3 0.4 0.4 0.3 0.2 0.3 0.3
0.0 0.1 0.2 0.3 0.3 0.3 0.2 0.2 0.3 0.5
0.0 0.0 0.2 0.2 0.4 0.3 0.2 0.1 0.3 0.5
0.0 0.0 0.2 0.2 0.4 0.2 0.1 0.0 0.1 0.4
79.6°W
79.3°W
79.1°W
78.8°W
78.6°W
78.4°W
78.1°W
Accumulated rain (mm)
64
56
48
40
32
24
16
8
0


t1
t2
t3
t4
t5
t6
t7
t8
t9
t10
t11
t12
Run 1
Run 2
Run 3
Run 4
Run 5
Run 6
Run 7
Run 8
Run 9
Run 10

Run 1 t1



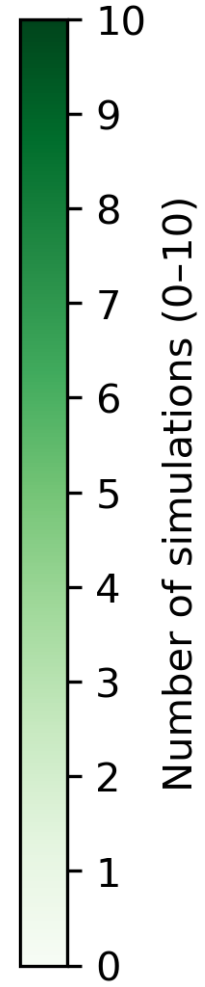
Condition: rain > 1.3 mm/h

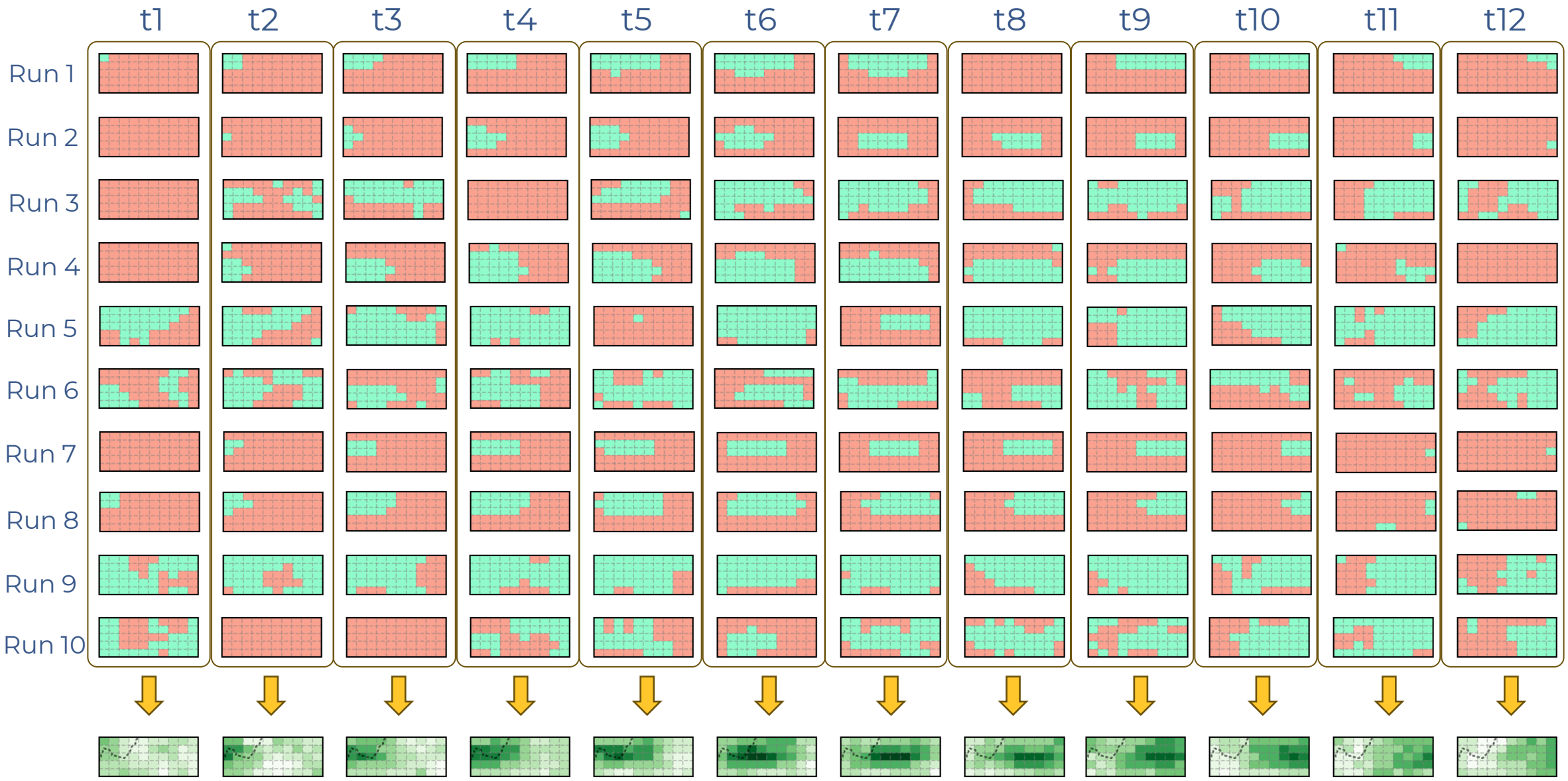
True/False rain > 1.3 mm/h



Number of runs that predicted rain > 1.3 mm/h → t1

5	4	2	1	1	2	2	3	3	2
4	4	2	1	1	3	4	3	1	2
4	4	2	2	1	2	2	3	1	1
3	3	3	2	2	2	1	2	2	1
3	3	4	3	3	2	0	2	3	1

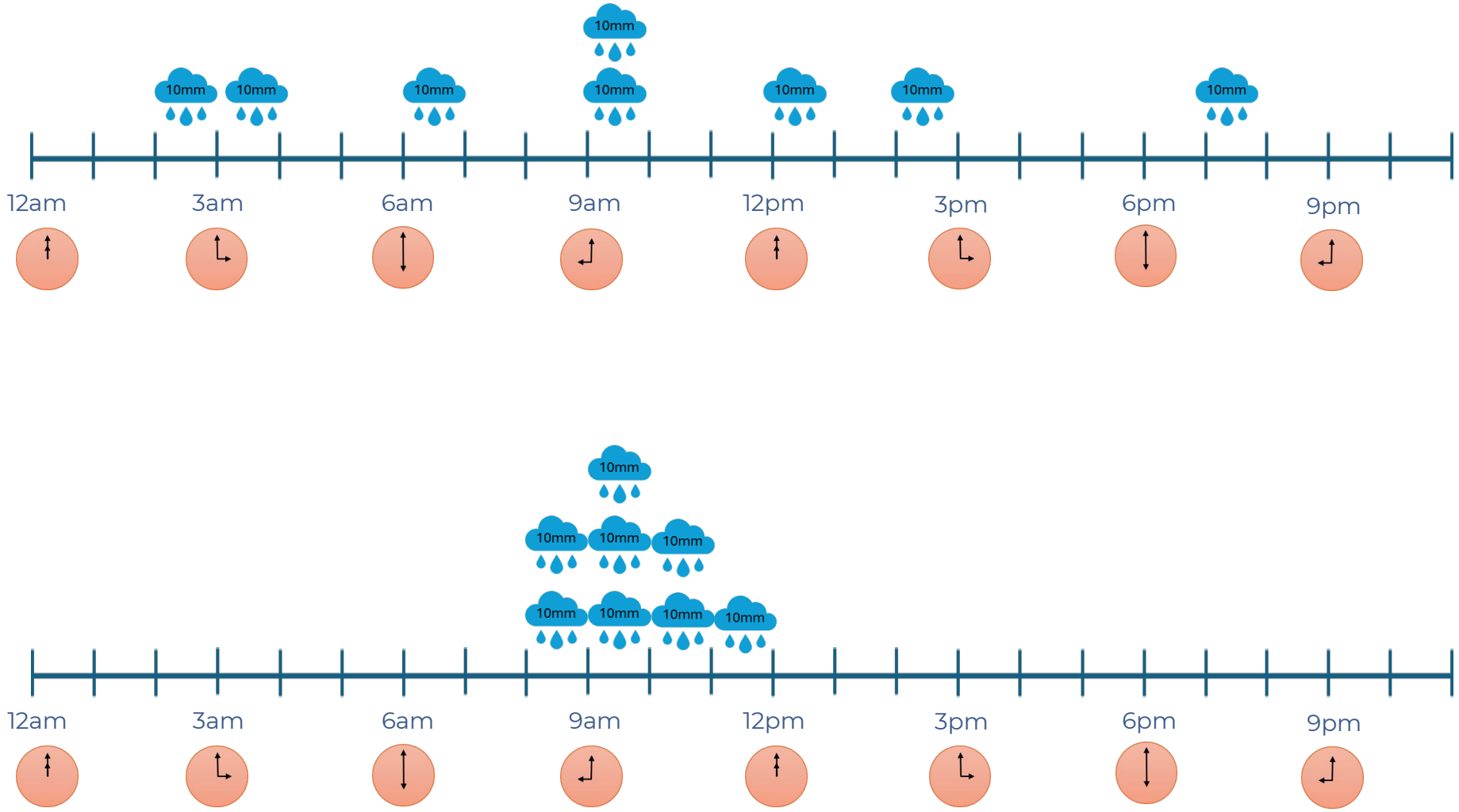




Number of runs that predicted rain > 1.3 mm/h in each time step

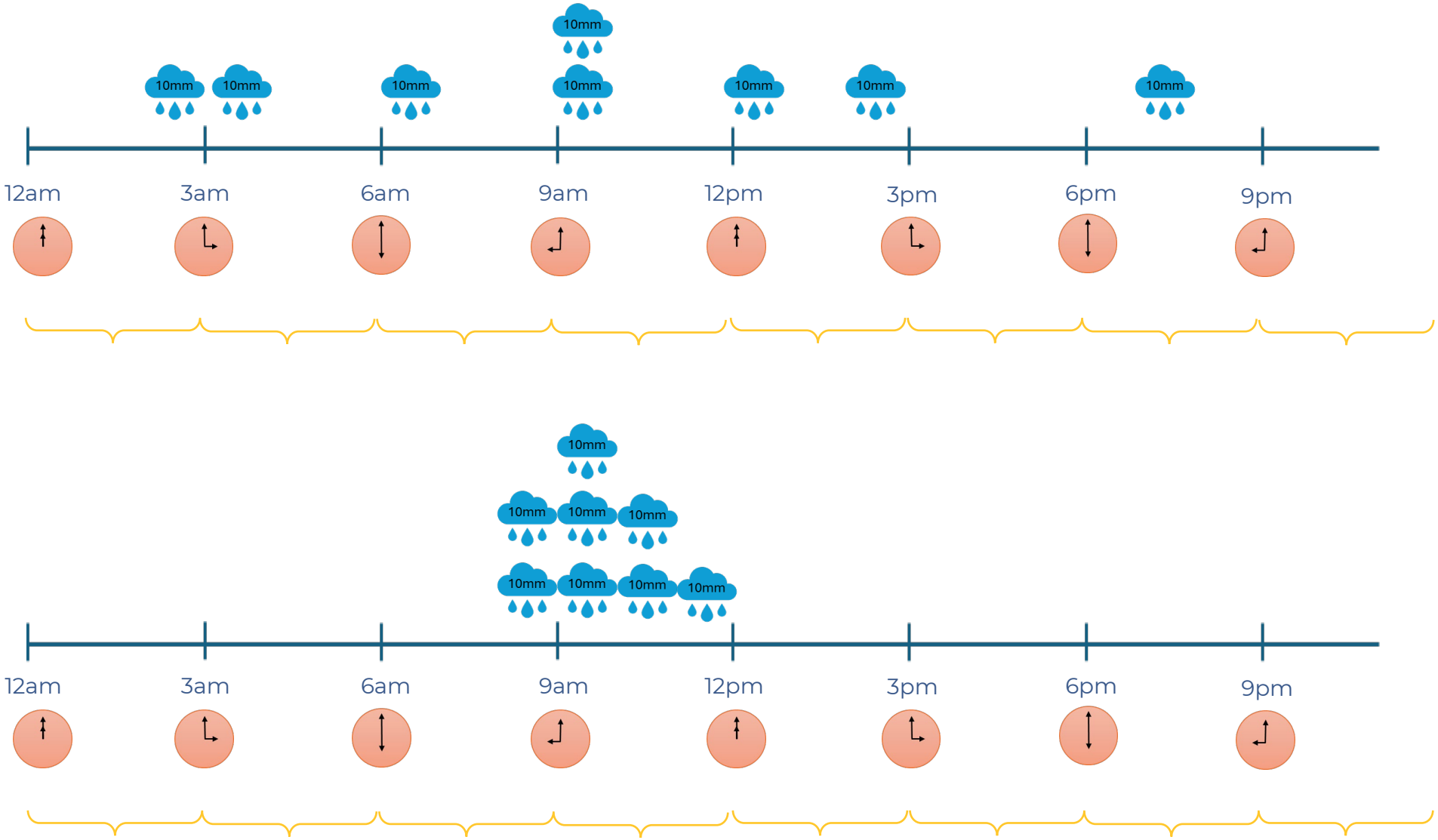
Variable

Rain



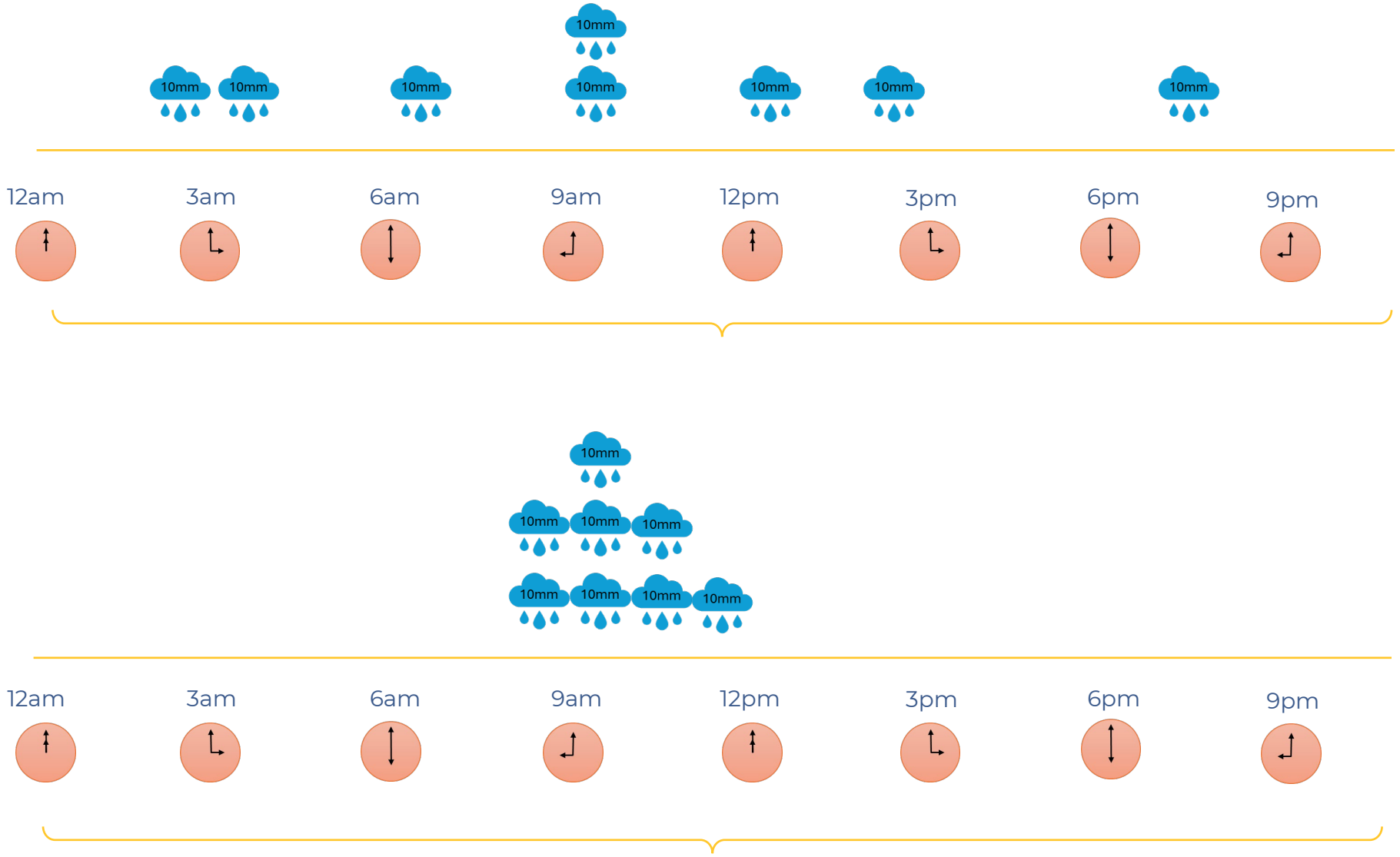
Variable

Rain



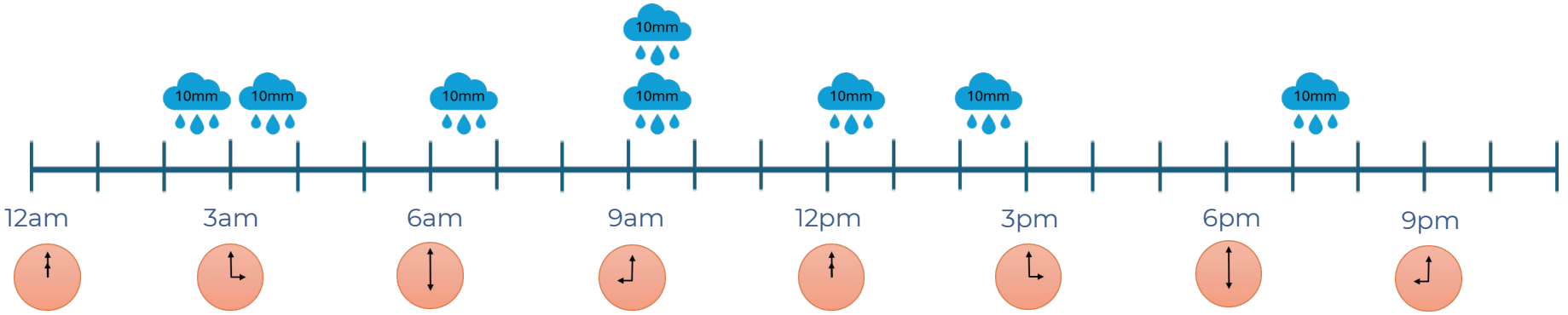
Variable

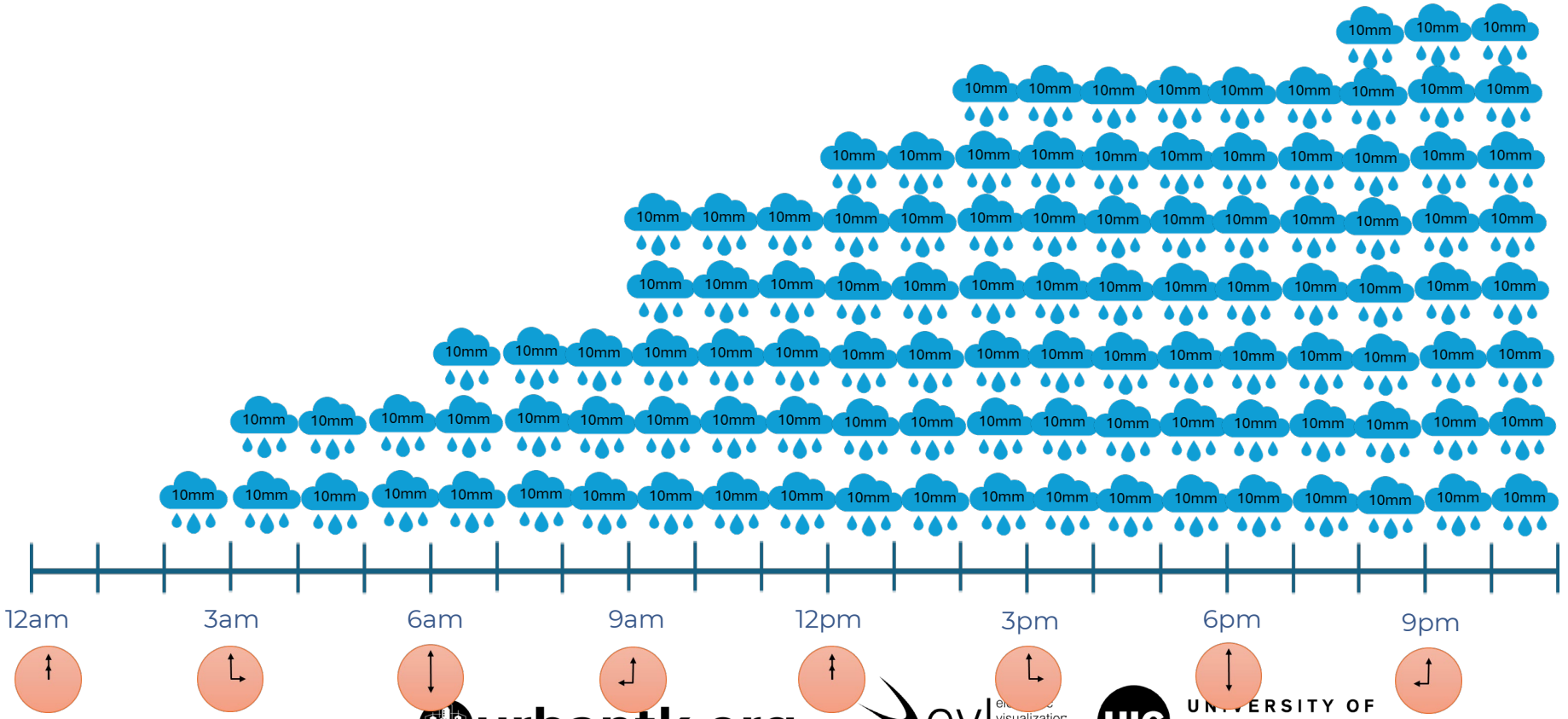
Rain



Variable

Rain

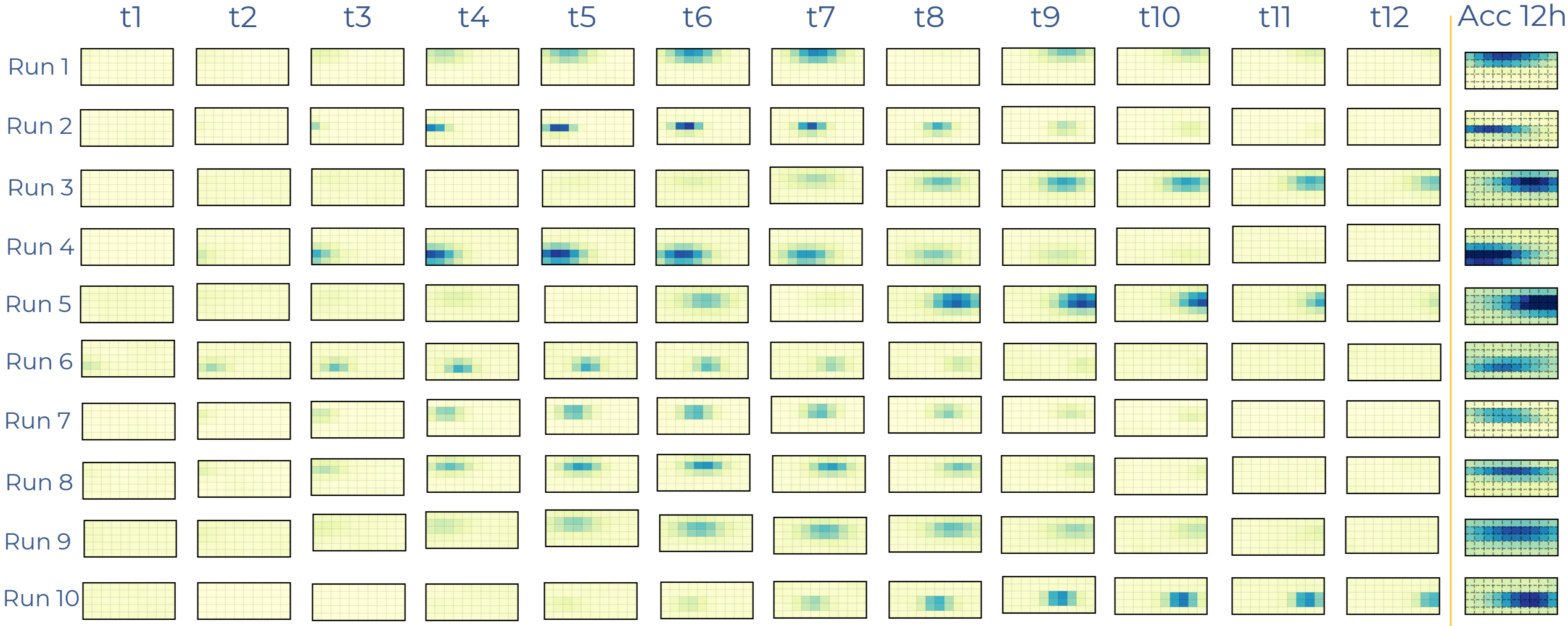


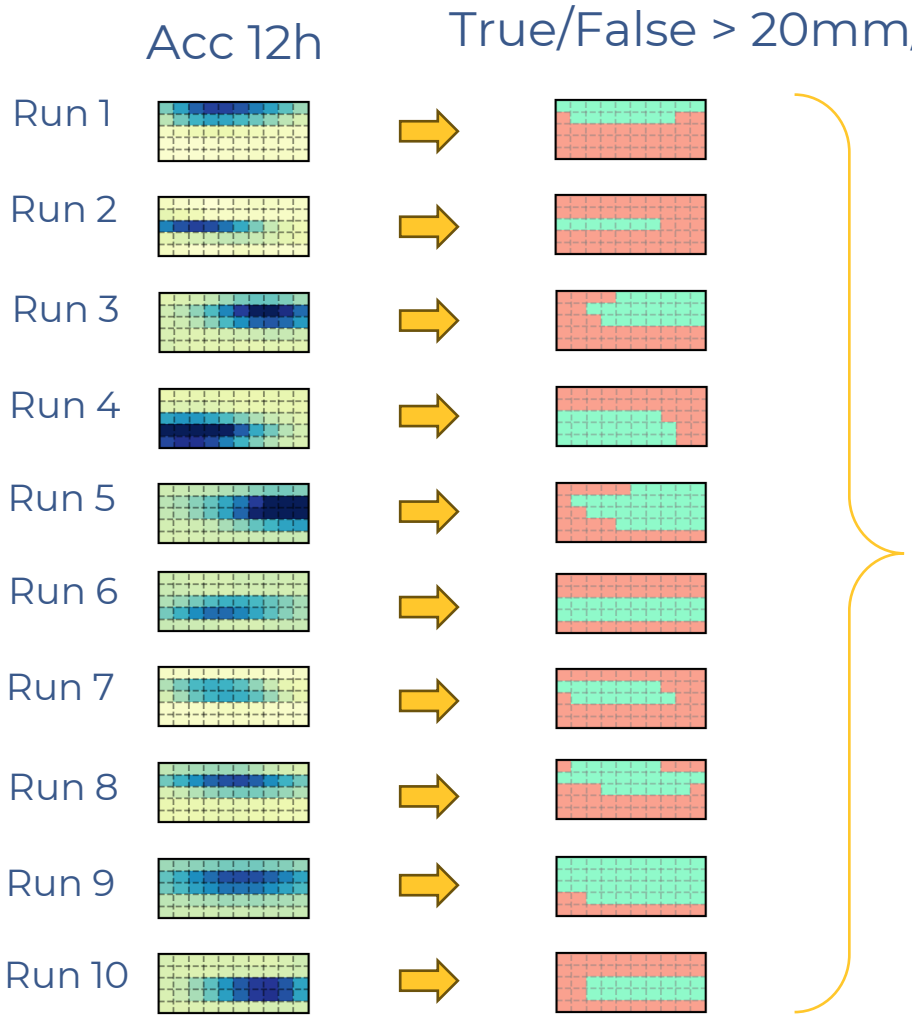


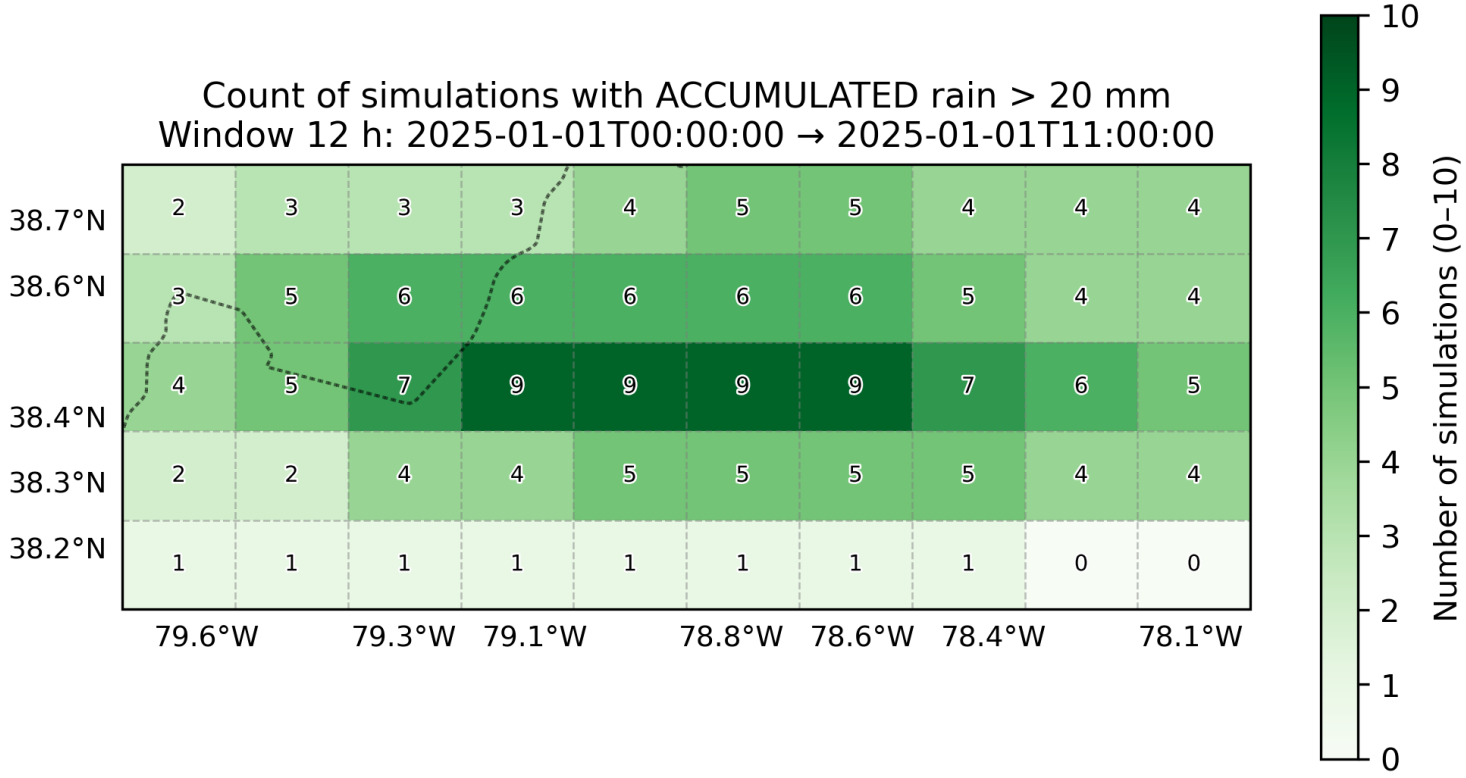
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t1
t2
t3
t4
t5
t6
t7
t8
t9
t10
t11
t12
Acc 12h
Run 1
Run 2
Run 3
Run 4
Run 5
Run 6
Run 7
Run 8
Run 9
Run 10

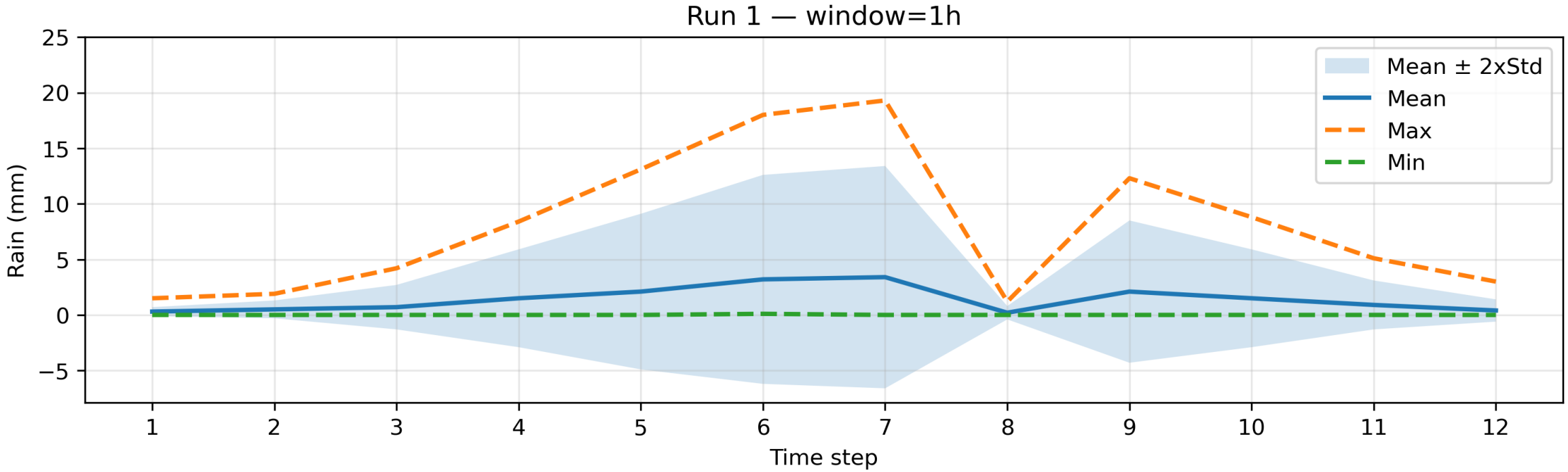

Acc 12h
True/False > 20mm/12h
Run 1
Run 2
Run 3
Run 4
Run 5
Run 6
Run 7
Run 8
Run 9
Run 10

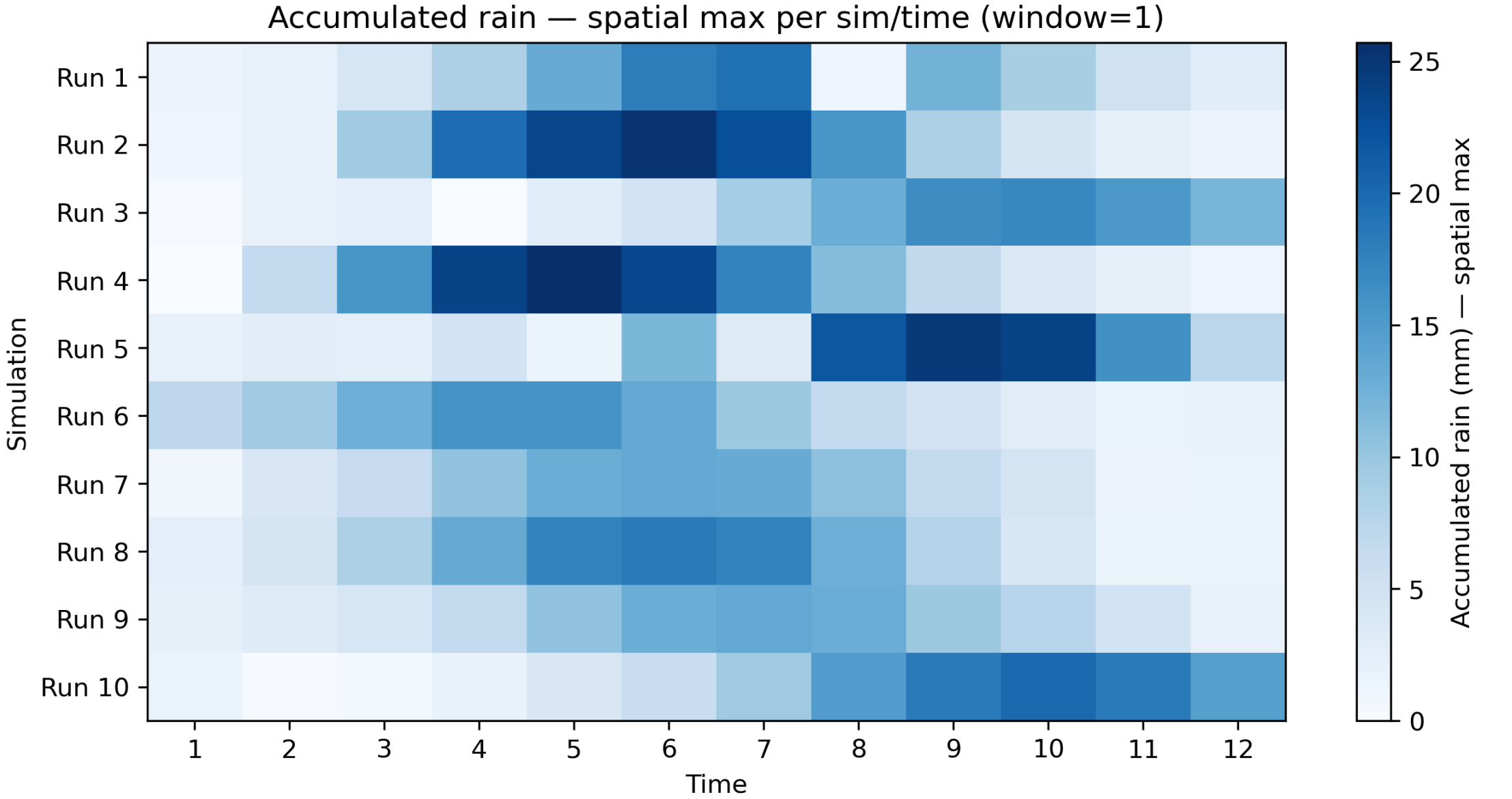

Count of simulations with ACCUMULATED rain > 20 mm
Window 12 h: 2025-01-01T00:00:00 → 2025-01-01T11:00:00
38.7°N
38.6°N
38.4°N
38.3°N
38.2°N
79.6°W
79.3°W
79.1°W
78.8°W
78.6°W
78.4°W
78.1°W
2 3 3 3 4 5 5 4 4 4
3 5 6 6 6 6 6 5 4 4
4 5 7 9 9 9 9 7 6 5
2 2 4 4 5 5 5 5 4 4
1 1 1 1 1 1 1 1 0 0
Number of simulations (0–10)
0 1 2 3 4 5 6 7 8 9 10

Which simulations had the highest predicted rainfall?

How much? Where?

	t1	t2	t3	t4	t5	t6	t7	t8	t9	t10	t11	t12
Run 1												
Avg	0.3	0.5	0.7	1.5	2.1	3.2	3.4	0.2	2.1	1.5	0.9	0.4
Sd	0.2	0.4	1.0	2.2	3.5	4.7	5.0	0.3	3.2	2.2	1.1	0.5
Max	1.5	1.9	0.0	8.4	13.1	18.0	19.3	1.2	12.3	8.8	5.1	3.0
Min	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0

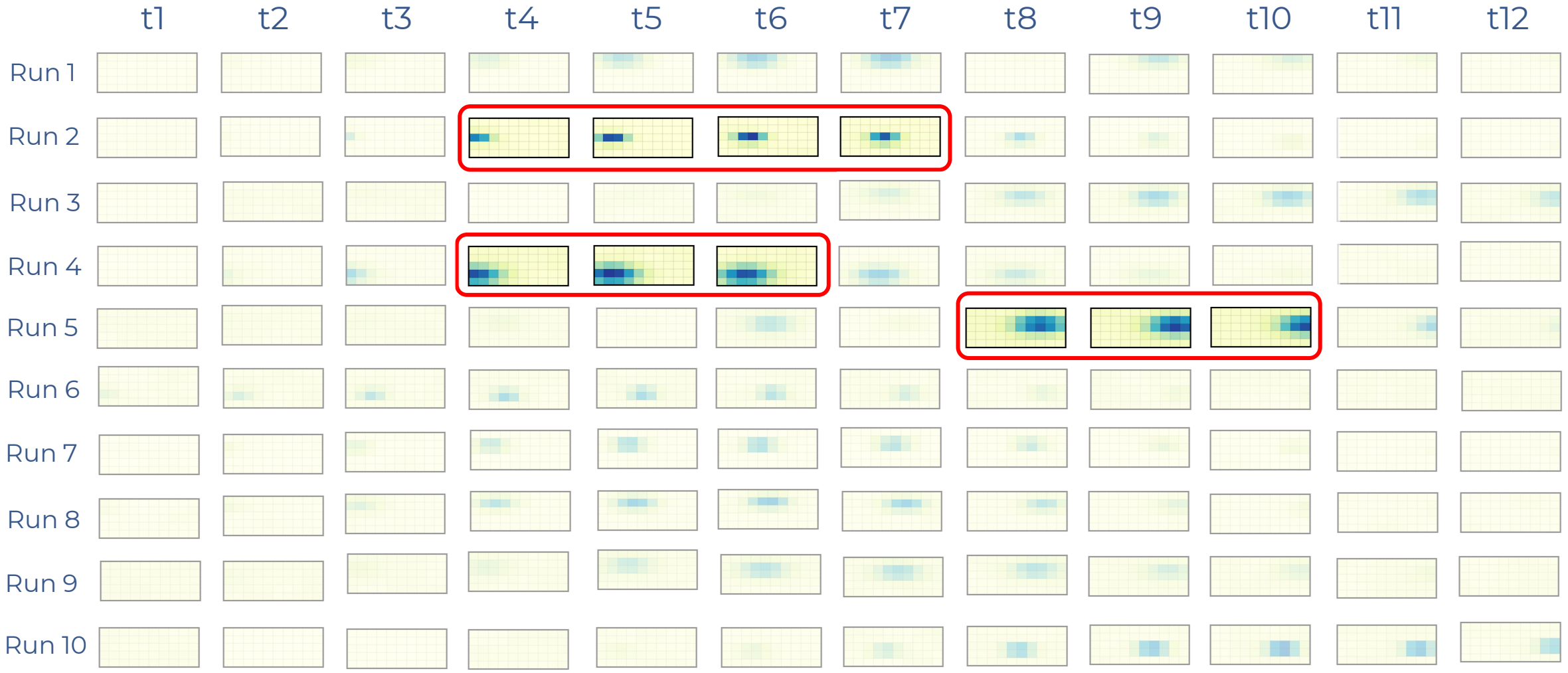




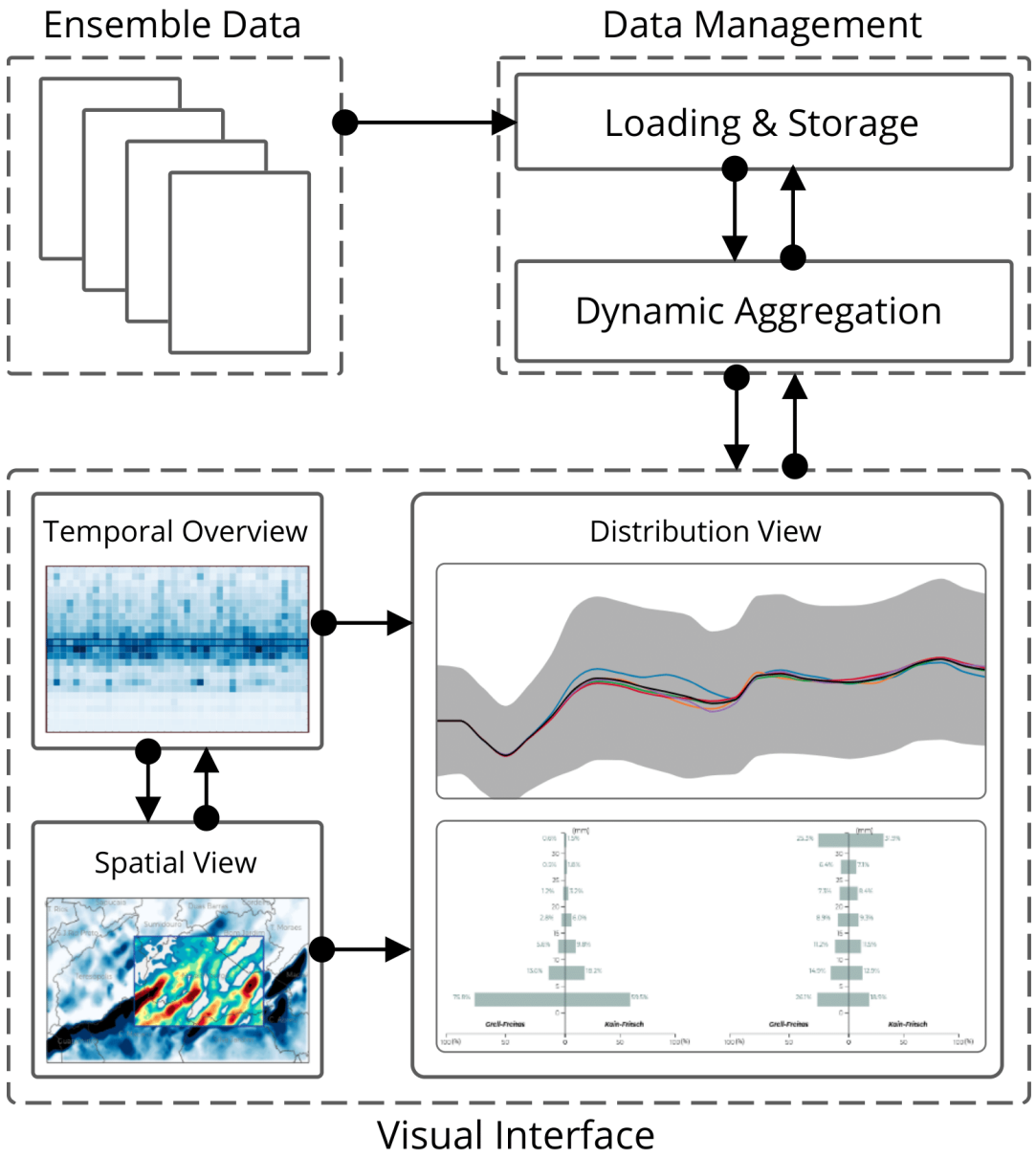
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t1
t2
t3
t4
t5
t6
t7
t8
t9
t10
t11
t12
Run 1
Run 2
Run 3
Run 4
Run 5
Run 6
Run 7
Run 8
Run 9
Run 10

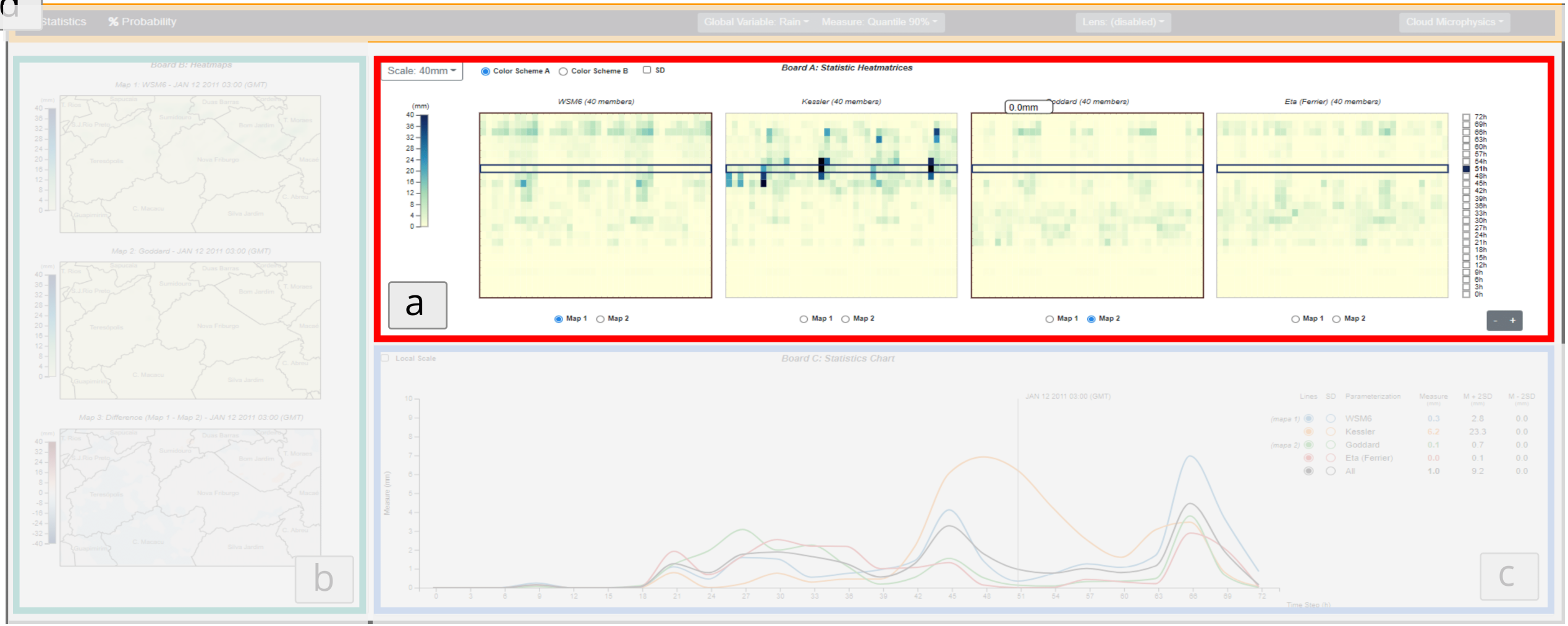
X-Weather


Ensemble Data
Data Management
Loading & Storage
Dynamic Aggregation
Temporal Overview
Distribution View
Spatial View
Visual Interface

d


Statistics
% Probability
Global Variable: Rain
Measure: Quantile 90%
Lens: (disabled)
Cloud Microphysics
Board B: Heatmaps
Map 1: WSM6 - JAN 12 2011 03:00 (GMT)
Map 2: Goddard - JAN 12 2011 03:00 (GMT)
Map 3: Difference (Map 1 - Map 2) - JAN 12 2011 03:00 (GMT)
b
Board A: Statistic Heatmatrices
Scale: 40mm
Color Scheme A
Color Scheme B
SD
WSM6 (40 members)
Kessler (40 members)
Goddard (40 members)
Eta (Ferrier) (40 members)
0.0mm
Map 1
Map 2
a
Board C: Statistics Chart
Local Scale
JAN 12 2011 03:00 (GMT)
Measure (mm)
Time Step (h)
Lines
SD
Parameterization
Measure
M + 2SD
M - 2SD
WSM6
Kessler
Goddard
Eta (Ferrier)
All
c

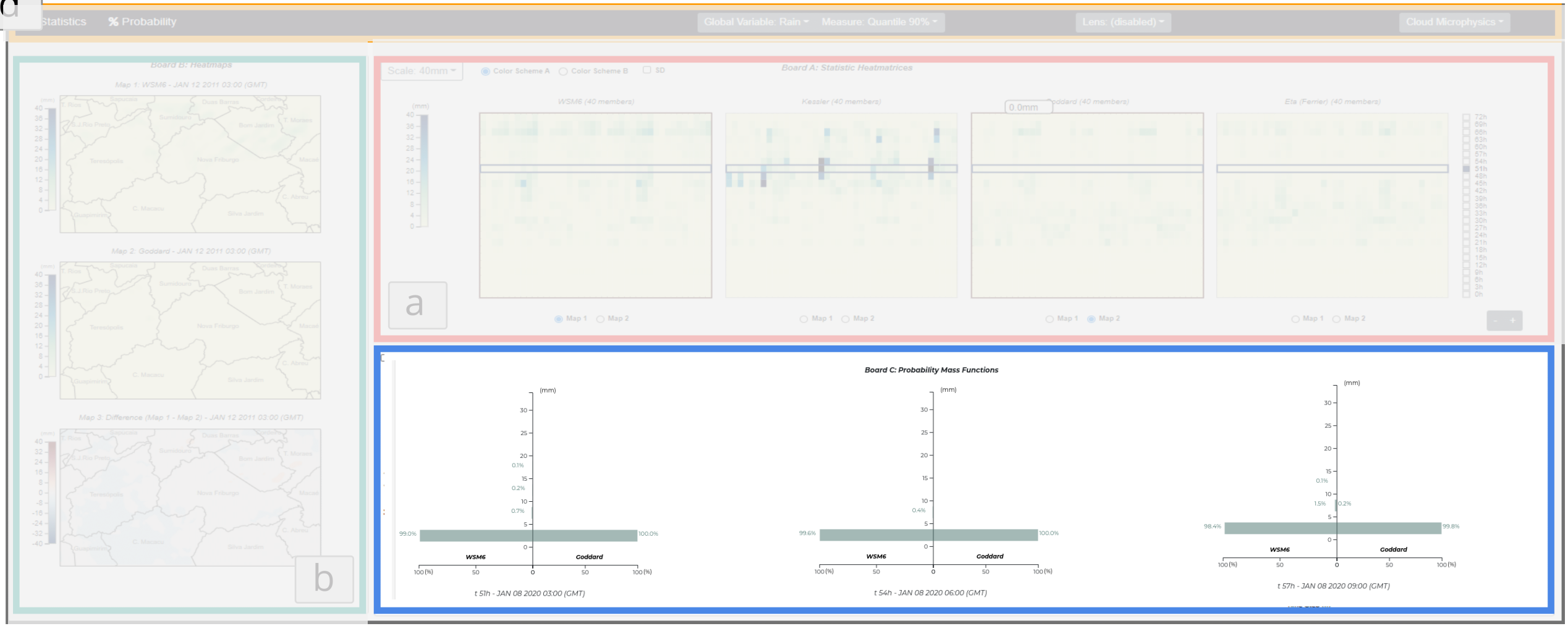
d

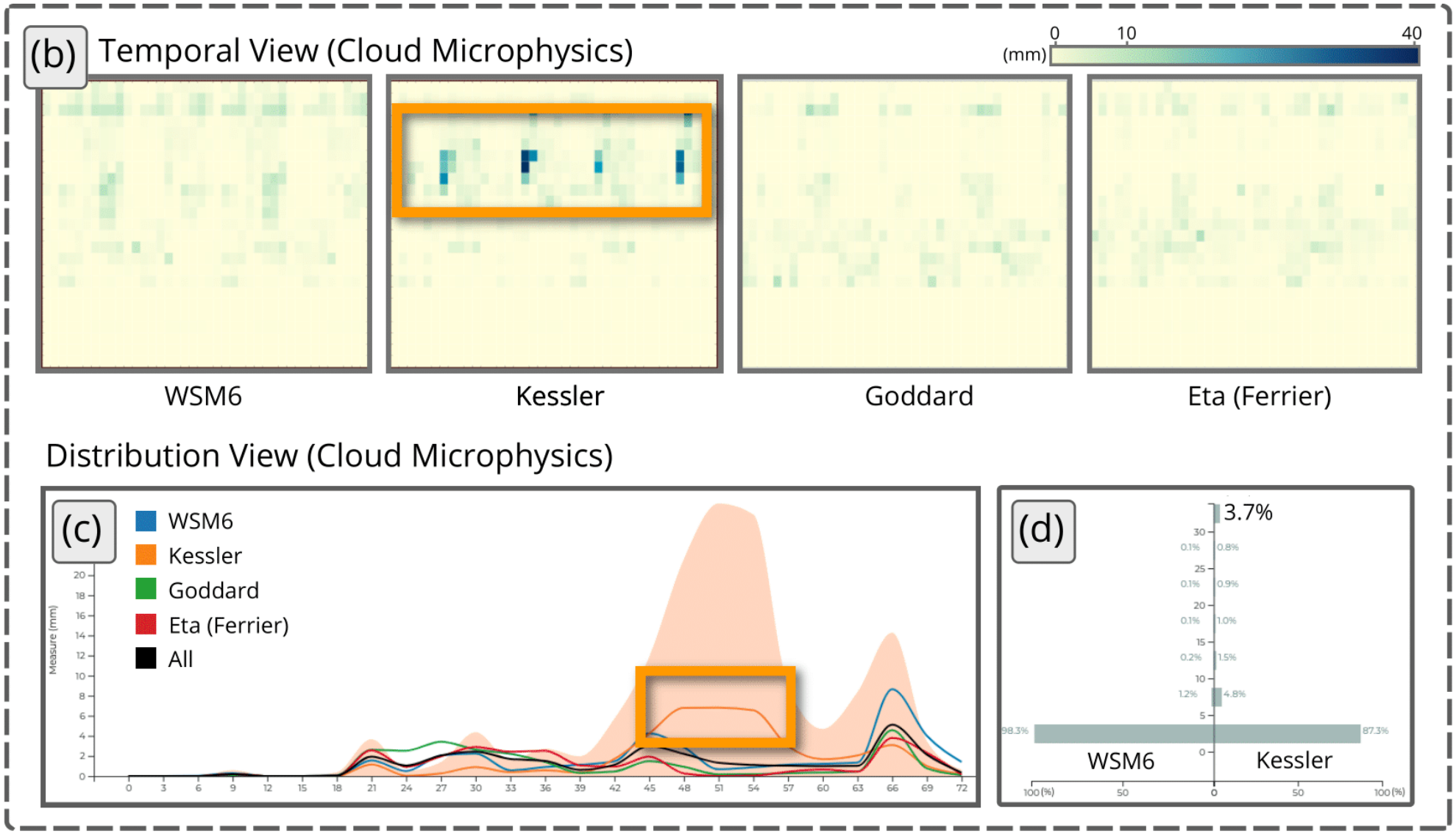

Statistics
% Probability
Global Variable: Rain
Measure: Quantile 90%
Lens: (disabled)
Cloud Microphysics
Board B: Heatmaps
Map 1: WSM6 - JAN 12 2011 03:00 (GMT)
Map 2: Goddard - JAN 12 2011 03:00 (GMT)
Map 3: Difference (Map 1 - Map 2) - JAN 12 2011 03:00 (GMT)
b
Scale: 40mm
Color Scheme A
Color Scheme B
SD
Board A: Statistic Heatmatrices
WSM6 (40 members)
Kessler (40 members)
0.0mm
Eta (Ferrier) (40 members)
(mm)
Map 1
Map 2
a
Local Scale
Board C: Statistics Chart
JAN 12 2011 03:00 (GMT)
Measure (mm)
Time Step (h)
Lines
SD
Parameterization
Measure
M + 2SD
M - 2SD
WSM6
Kessler
Goddard
Eta (Ferrier)
All
c

d


Statistics
% Probability
Global Variable: Rain
Measure: Quantile 90%
Lens: (disabled)
Cloud Microphysics
Map 1: WSM6 - JAN 12 2011 09:00 (GMT)
Map 2: Kessler - JAN 12 2011 09:00 (GMT)
Map 3: Difference (Map 1 - Map 2) - JAN 12 2011 09:00 (GMT)
Sapucaia
Duas Barras
Cordeiro
T. Rios
S.J.Rio Preto
Sumidouro
Bom Jardim
T. Moraes
Teresópolis
Nova Friburgo
Macaé
C. Abreu
C. Macacu
Guapimirim
Silva Jardim
Board A: Statistic Heatmatrices
Scale: 40mm
Color Scheme A
Color Scheme B
SD
WSM6 (40 members)
Kessler (40 members)
Goddard (40 members)
Eta (Ferrier) (40 members)
0.0mm
Map 1
Map 2
a
Board C: Statistics Chart
Local Scale
JAN 12 2011 03:00 (GMT)
Measure (mm)
Time Step (h)
Lines
SD
Parameterization
Measure
M + 2SD
M - 2SD
WSM6
Kessler
Goddard
Eta (Ferrier)
All
c

d


Statistics
% Probability
Global Variable: Rain
Measure: Quantile 90%
Lens: (disabled)
Cloud Microphysics
Board B: Heatmaps
Map 1: WSM6 - JAN 12 2011 03:00 (GMT)
Map 2: Goddard - JAN 12 2011 03:00 (GMT)
Map 3: Difference (Map 1 - Map 2) - JAN 12 2011 03:00 (GMT)
b
Scale: 40mm
Color Scheme A
Color Scheme B
SD
Board A: Statistic Heatmatrices
WSM6 (40 members)
Kessler (40 members)
0.0mm
Eta (Ferrier) (40 members)
a
Map 1
Map 2
Board C: Probability Mass Functions
(mm)
99.0%
100.0%
WSM6
Goddard
t 51h - JAN 08 2020 03:00 (GMT)
99.6%
0.4%
t 54h - JAN 08 2020 06:00 (GMT)
98.4%
99.8%
1.5%
0.2%
t 57h - JAN 08 2020 09:00 (GMT)



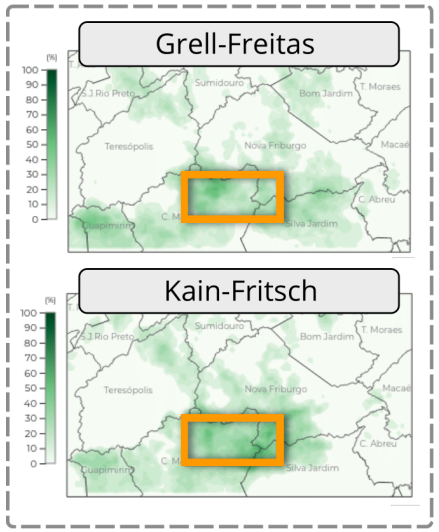
Accumulated Rain

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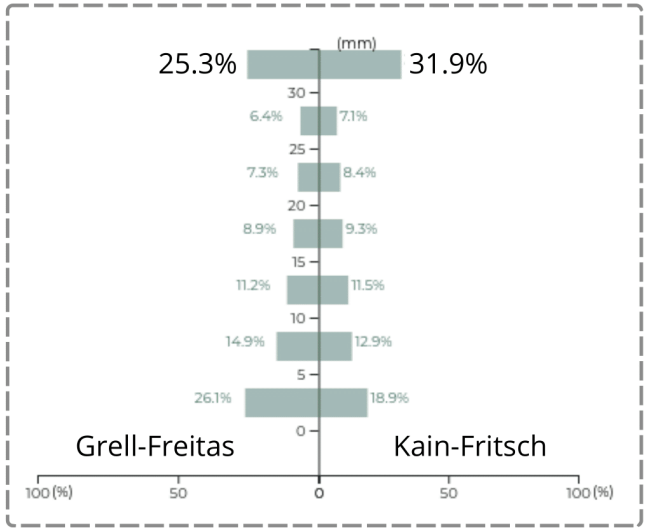
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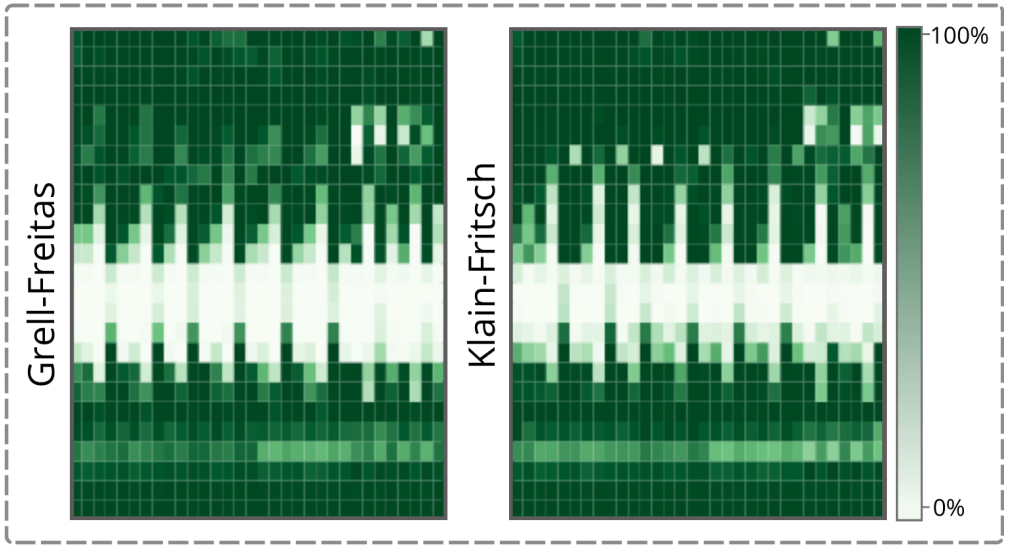
(a) P(Rain >= 30mm)



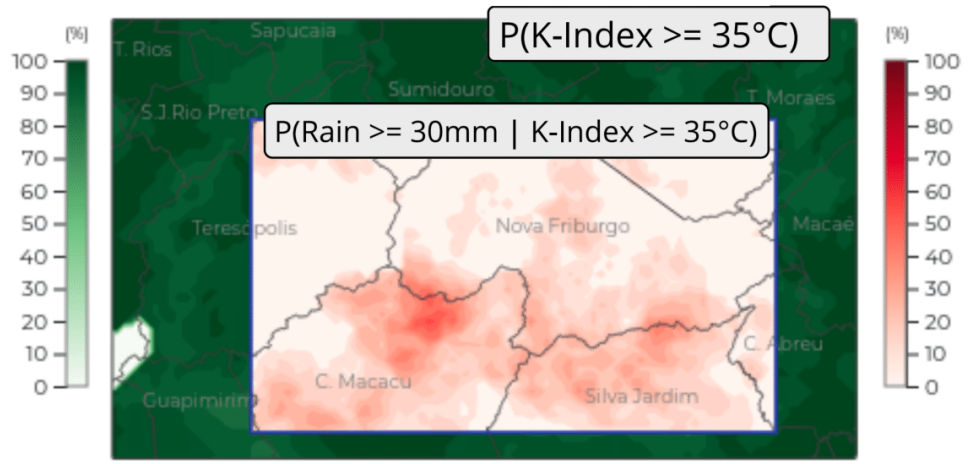
(b) Accumulated Rain Probability



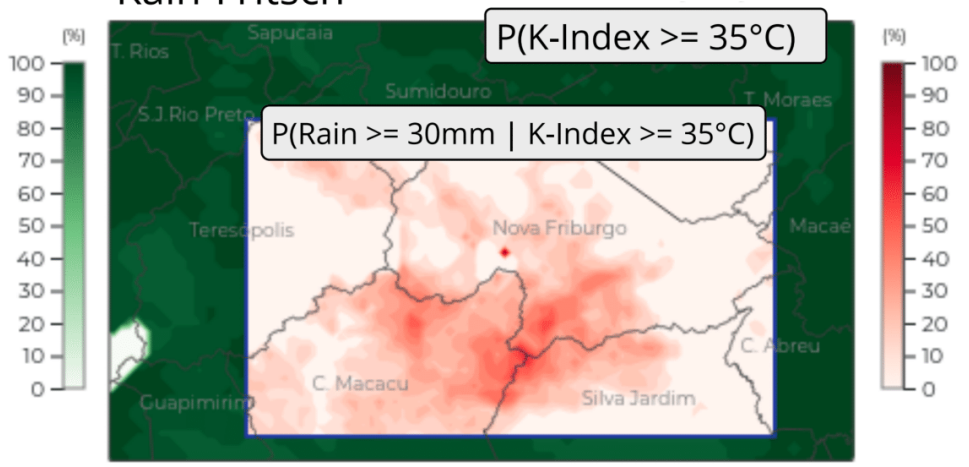
(c) P(K-Index >= 35 °C)



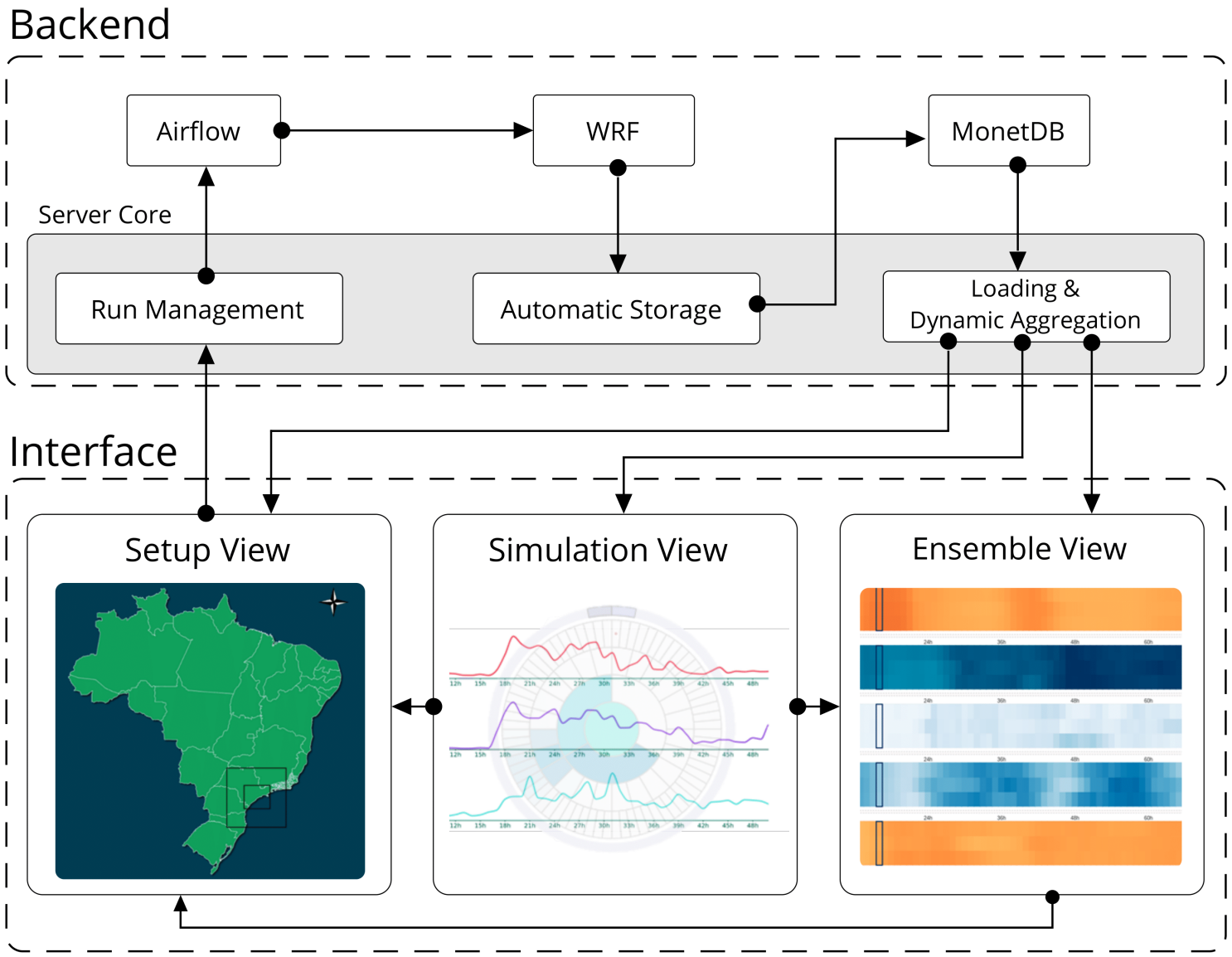
Grell-Freitas



Kain-Fritsch



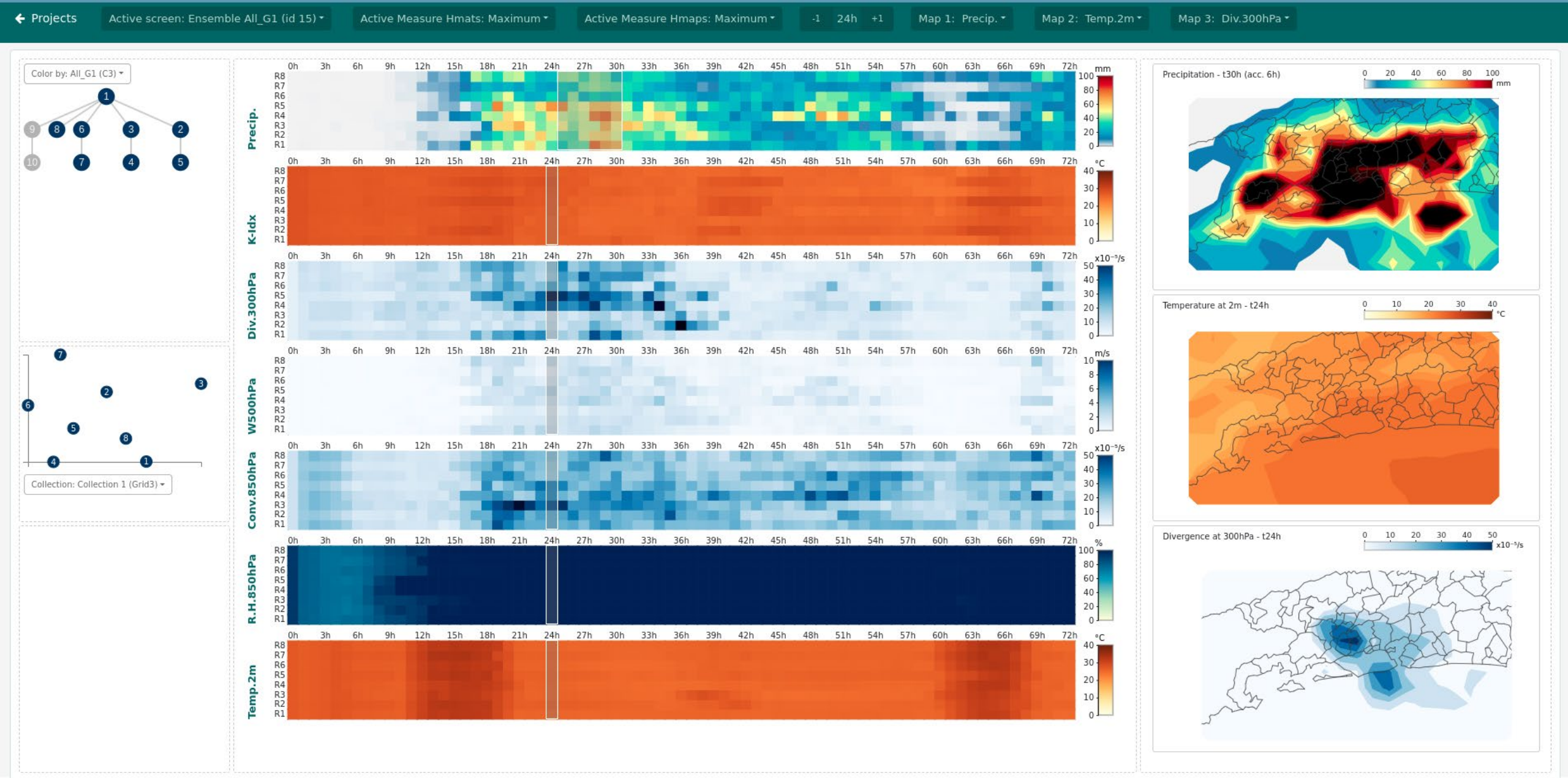
ProWis

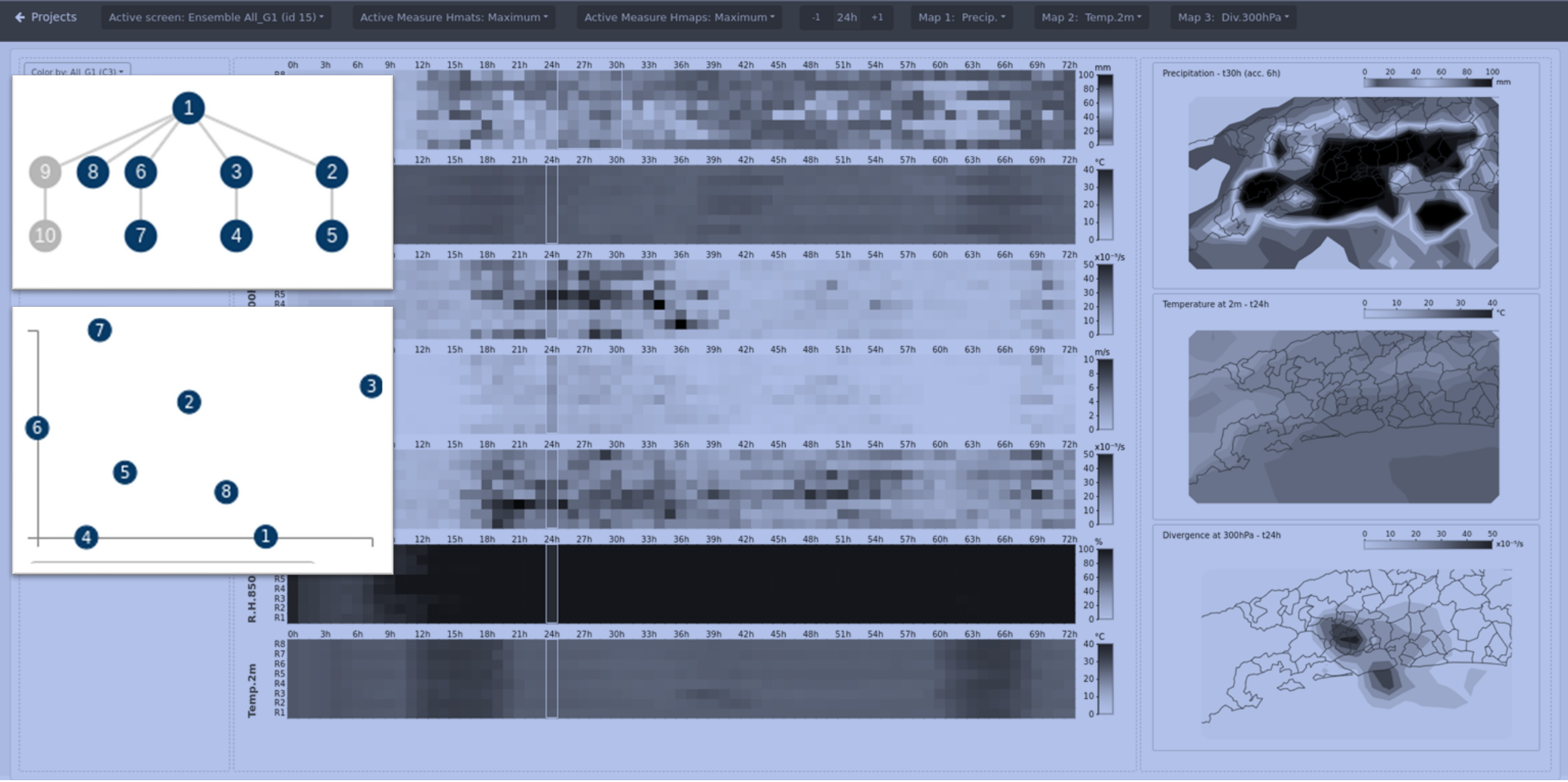


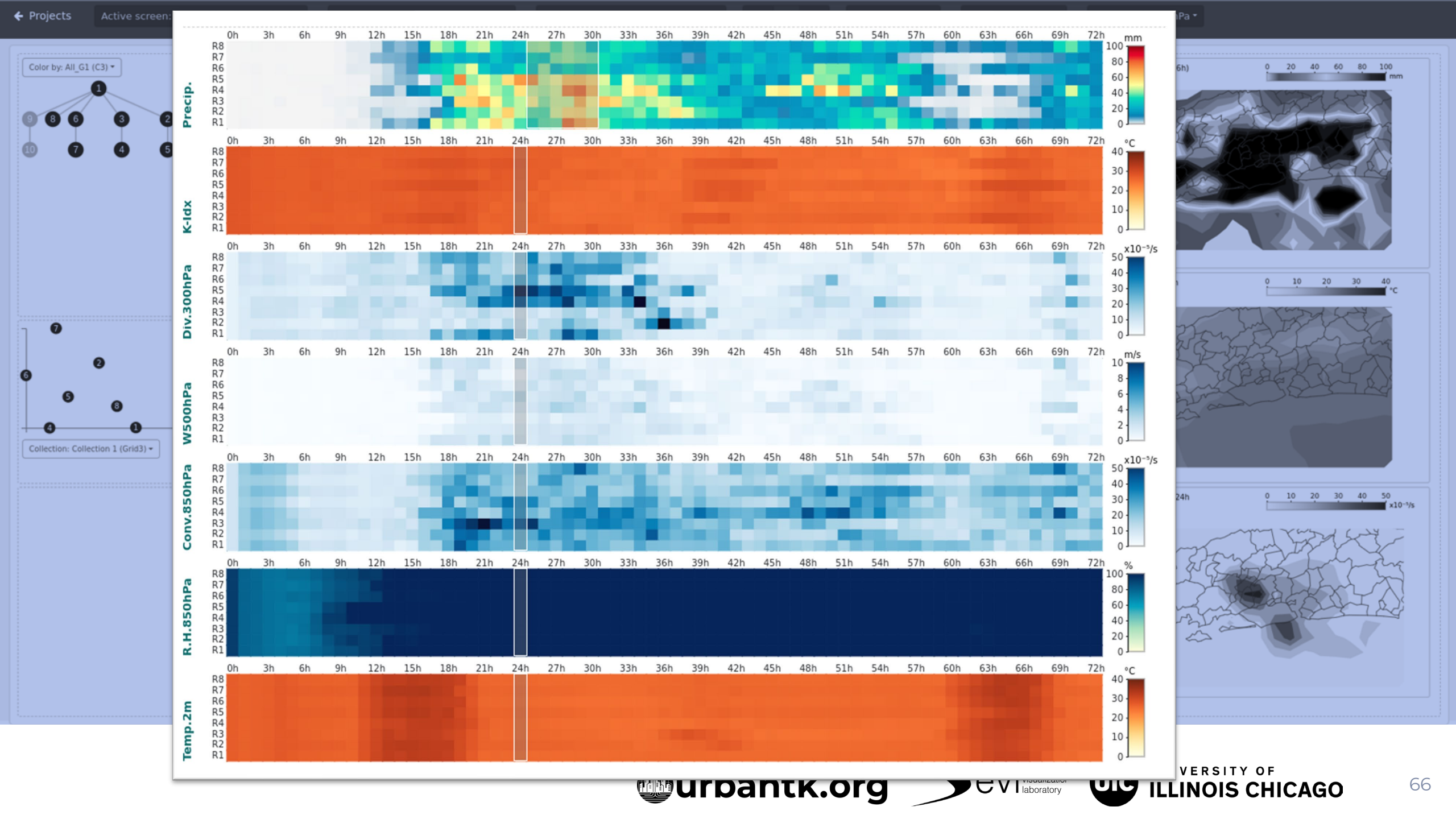
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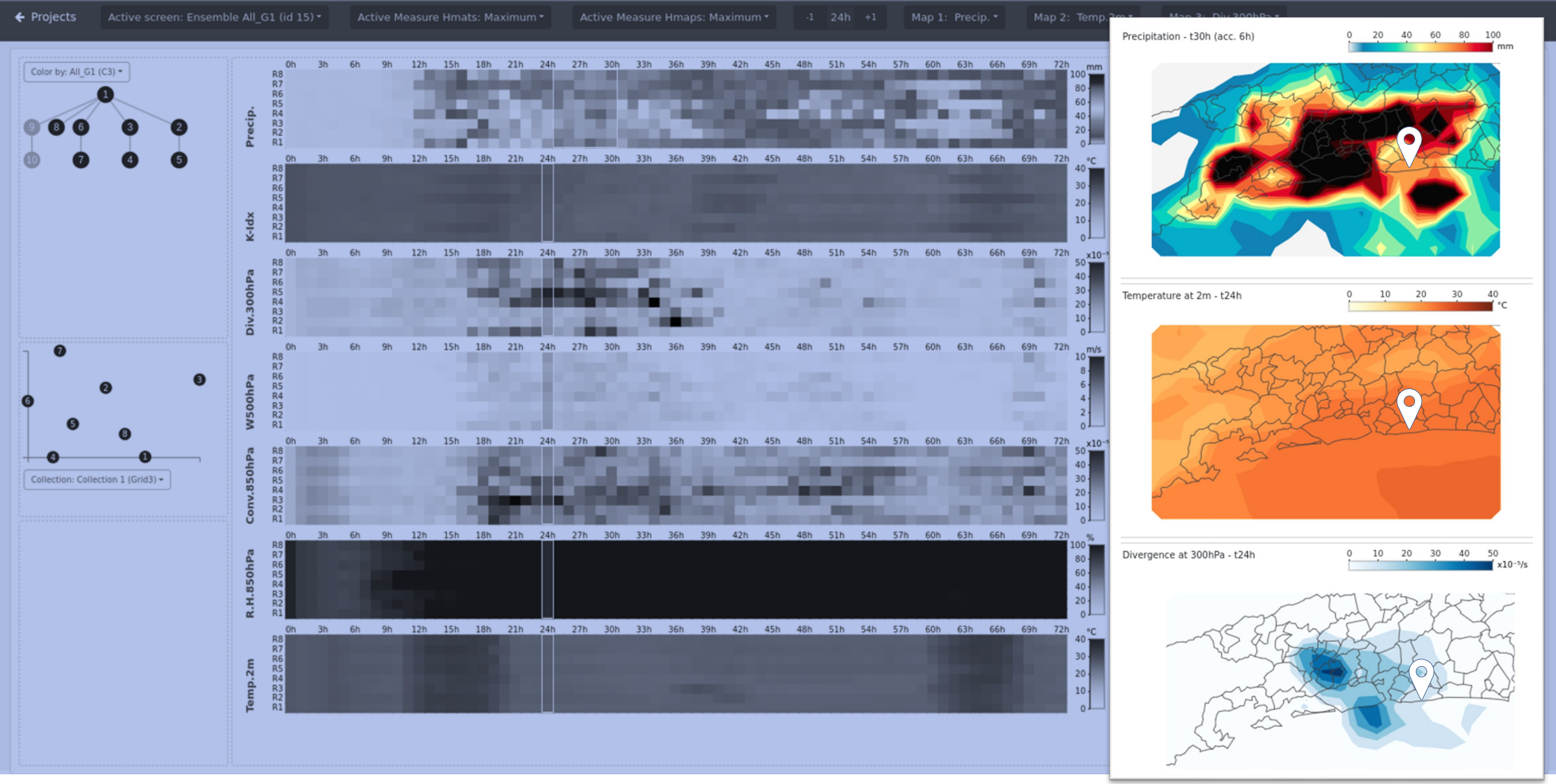
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UIC UNIVERSITY OF ILLINOIS CHICAGO


Projects
Active screen: Ensemble All_G1 (id 15)
Active Measure Hmats: Maximum
Active Measure Hmaps: Maximum
-1 24h +1
Map 1: Precip.
Map 2: Temp.2m
Map 3: Div.300hPa
Color by: All_G1 (C3)
Collection: Collection 1 (Grid3)
Precip.
K-Idx
Div.300hPa
W500hPa
Conv.850hPa
R.H.850hPa
Temp.2m
Precipitation - t30h (acc. 6h)
Temperature at 2m - t24h
Divergence at 300hPa - t24h


Projects
Active screen: Ensemble All_G1 (id 15)
Active Measure Hmats: Maximum
Active Measure Hmaps: Maximum
-1 24h +1
Map 1: Precip.
Map 2: Temp.2m
Map 3: Div.300hPa
Color by: All_G1 (C3)
1
9 8 6 3 2
10 7 4 5
7
3
2
6
5
8
4 1
R.H.850
Temp.2m
Precipitation - t30h (acc. 6h)
mm
Temperature at 2m - t24h
°C
Divergence at 300hPa - t24h
x10⁻⁵/s


Projects
Active screen:
Color by: All_G1 (C3)
Collection: Collection 1 (Grid3)
Precip.
K-Idx
Div.300hPa
W500hPa
Conv.850hPa
R.H.850hPa
Temp.2m
0h 3h 6h 9h 12h 15h 18h 21h 24h 27h 30h 33h 36h 39h 42h 45h 48h 51h 54h 57h 60h 63h 66h 69h 72h
R8 R7 R6 R5 R4 R3 R2 R1
mm
°C
x10⁻⁵/s
m/s
%


Projects
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Active Measure Hmats: Maximum
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°C
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x10⁻⁵/s